

IMPERIAL

MODULO ADC ACQUISITION FOR ULTRASOUND RADIO-FREQUENCY SIGNALS

AN OPERATING-ENVELOPE STUDY OF THE UNLIMITED SENSING
FRAMEWORK

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A Thesis submitted in fulfillment of requirements for the degree of
Master of Science in Sensor Systems and Networks

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Imperial College London
2026

Abstract

Capturing the full dynamic range of an ultrasound Radio-Frequency (RF) signal presents a major challenge in analogue-to-digital conversion, as the converter's input range must accommodate both the strongest and weakest echoes within a record to prevent information loss. The Unlimited Sensing Framework (USF) addresses this limitation by folding high-dynamic-range signals into a bounded interval $[-\lambda, \lambda)$ using a modulo Analogue-to-Digital Converter (ADC), subsequently reconstructing the original waveform through digital processing. This work investigates in simulation the recovery of ultrasound RF signals from centred-modulo measurements under realistic receive-chain conditions. The evaluation models key hardware non-idealities, including electronic noise on each receive channel, finite-bit quantisation, and variations in the wrap height of the modulo ADC. The proposed recovery algorithm estimates an initial fold sequence directly from the threshold λ and the modulo samples, validates this hypothesis against out-of-band spectral content, and executes, when needed, a targeted search over closely spaced, opposite-sign fold pairs generated by the ultrasound pulse geometry. The pipeline was evaluated on 5 MHz linear-array records simulated in Field II. Results demonstrate exact fold recovery wherever Itoh's condition holds, that is, wherever the largest step between consecutive samples of the original signal stays below the bounding amplitude of the modulo ADC, λ . Across the test records, this condition accommodated signal peaks up to 3λ , yielding a threefold dynamic range extension. Robustness trials evaluating channel noise, quantisation depth, and wrap error independently and in combination confirmed that recovery tolerance to hardware impairments is governed by the margin between λ and the maximum sample step. Finally, applying the folding and recovery pipeline across all channels of a full ultrasound scan, successfully reproduced a reference B-mode image, whereas a conventional saturating converter of equivalent input range caused image corruption.

Plain Language Summary

An ultrasound scanner has to listen for echoes of a wide range of amplitudes at the same time. Some reflections, from bone or the wall of an organ, are very loud. Others, from blood or the fine texture of soft tissue, are very quiet. A useful analogy is like a camera that must capture a bright lamp and a dim shadow in a single shot. Scanners use a sensor with a fixed loudness limit. Anything louder than that limit is cut off, and the loudest parts of the echo are lost. Turning the sensitivity down to protect the loud echoes then hides the quiet ones.

This project tests a different kind of sensor, in computer simulations. Instead of cutting off a loud echo, the sensor wraps it back into its allowed range, the way a clock hand wraps from 12 back to 1. A calculation then works out how many times the echo wrapped and when, and unfolds it, thus recovering the full signal. The study asked how far the sensor's limit can be lowered before recovering the full signal stops working, and how much electronic noise and imperfect wrapping it can tolerate.

On simulated scans of a tissue-like scene, the unfolding recovered the original echoes exactly up to a predictable limit set by how quickly the waveform changes from one instant to the next. At that limit the loudest echoes were about three times larger than the sensor's range. The same method, applied on every listening element of the probe, produced images that matched a perfect sensor. A conventional sensor with the same range left clear damage in the picture.

These results come from one simulated probe and one simulated scene. The wrapping threshold was assumed known exactly, and no real patient recordings were used. It nevertheless sets a precedent, in simulation, by showing the idea works.

Declaration of Originality

I hereby declare that the work presented in this thesis is my own unless otherwise stated. To the best of my knowledge the work is original and ideas developed in collaboration with others have been appropriately referenced.

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List of Acronyms

ADC	Analogue-to-Digital Converter
DAS	Delay-and-Sum
DFT	Discrete Fourier Transform
US	Ultrasound
NRMSE	Normalised Root-Mean-Square Error
RF	Radio-Frequency
SNR	Signal-to-Noise Ratio
SVD	Singular Value Decomposition
USF	Unlimited Sensing Framework

List of Symbols

Accents and general conventions

$\hat{x}$	estimate of x produced by the recovery
$\mathbf{x}$	vector (bold); scalars plain; sets calligraphic

Signal model and sampling

$g[k]$	ultrasound RF record (the signal to be unfolded)
$g_n[k]$	same record after additive channel noise
T_s	sampling period
f_c	centre frequency
$K; k$	number of samples; sample index
λ	modulo (folding) threshold
$\mathcal{M}_\lambda(\cdot)$	centred modulo operator, folding into $[-\lambda, \lambda)$
$y[k]$	folded (modulo) samples, $y[k] = \mathcal{M}_\lambda(g[k])$
FF	fold fraction, $\lambda = \text{FF} \cdot \max g $
B	converter resolution in bits

Difference and Fourier domain

Δ	first-difference operator, $(\Delta x)[k] = x[k+1] - x[k]$
$r[k]$	fold sequence (residual, staircase), $g = y + r$
$q; \hat{q}$	per-position staircase jump in units of 2λ ($\Delta r = 2\lambda q$); its direct read from the data
P	reserved harmonic count (in-band DFT bins occupied by the RF pulse)
$\mathbf{z}$	out-of-band DFT coefficients of $\Delta \mathbf{y}$

Pair recovery

$M; \widetilde{M}$	fold count; its estimate from the direct read
n_p	number of closed fold pairs, $n_p = \widetilde{M}/2$
$(p, w); s$	catalogue pair (start index, width); sign $s \in \{\pm 1\}$
$\hat{\mathbf{q}}_{\text{seed}}$	jump sequence implied by the direct-read seed
ρ	out-of-band misfit of a candidate jump sequence
$\mathcal{G}$	gated candidate fold indices
$\mathcal{C}$	pair catalogue built on the gate
$\mathcal{Q}$	candidate fold sequences formed from signed pairs in $\mathcal{C}$
$\eta_g; \eta_j$	gate fractions of λ : rail cue; jump cue
d	gate dilation half-width [samples]
w_{max}	maximum pair width [samples]
θ	seed acceptance tolerance ($\rho \leq \theta \ \mathbf{z}\ _2$)
W	number of partial candidates kept per search stage

Operating regime

m	read margin, $m = \lambda - \max \Delta g $
FF _{bound}	Itoh floor as a fold fraction, $\max \Delta g / \max g $

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1

Introduction

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1.1 Context of the Study

Medical Ultrasound (US) forms an image of the body by transmitting a short acoustic pulse into tissue and recording the echoes that return to the probe. The recorded echo trace, known as the Radio-Frequency (RF) signal, contains within a single record both very strong and very weak reflections, for instance, the interface between soft tissue and an organ or bone returns almost the entire incident pulse whereas blood and the fine texture of soft tissue return only a fraction of it. This produces a signal with a wide dynamic range. For instance, along a single beam path through the heart, the strongest and the weakest echoes of clinical interest are separated by roughly 70 dB [1, Sec. 10.8.2], and a receiver that is to detect the faint echoes without saturating on the strong ones must accommodate that entire span.

Before any image can be formed, the RF signal on each receive channel passes through an

1

Analogue-to-Digital Converter (ADC), which accepts a limited input range. Any part of the waveform that exceeds that range is replaced by the rail value, so that the amplitude information above the rail is thrown away or "clipped", and permanently lost (see Fig. 1.1 (a)–(b)). Widening the input range at a fixed bit depth coarsens the quantisation step, and reducing the gain so that the brightest echo fits inside the range pushes the weakest echoes towards the quantisation floor, so in either case, the converter is forced to sacrifice one end of the span in order to keep the other. The input range of the converter therefore constitutes a bottleneck on the dynamic range that can be captured in the image.

The Unlimited Sensing Framework (USF) [2] is a signal acquisition scheme that circumvents clipping by employing a modulo ADC that folds any excursion beyond its range back into the interval $[-\lambda, \lambda)$ (Fig. 1.1 (c)). Consequently, the recorded samples remain bounded regardless of the input amplitude, preserving the complete waveform without information loss. The framework subsequently reconstructs the signal by algorithmically deducing the precise timing, direction, and total count of the fold events. This information forms a piecewise-constant sequence, referred to throughout this work as the fold sequence r . Because each folding event shifts the signal by twice the converter threshold λ , every value in r is an integer multiple of 2λ . Adding r back to the modulo samples restores the original record (Fig. 1.1 (d)). Unfolding an RF line therefore reduces to recovering r solely from the modulo measurements; evaluating whether this recovery can be achieved reliably for ultrasound RF signals is the central objective of this work.

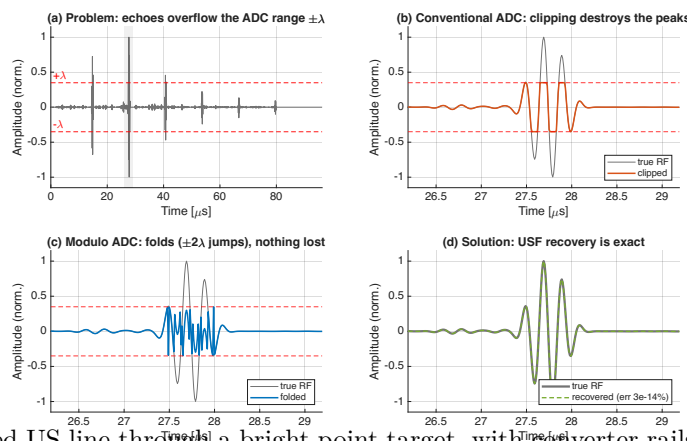


Figure 1.1: Simulated US line through a bright point target, with converter rails $\pm\lambda$ covering 35% of the peak. (a) Echoes exceed the rails. The shaded window marks the brightest of them. (b) A conventional ADC clips those peaks. (c) A modulo ADC wraps the same peaks back into $[-\lambda, \lambda)$ by jumps of height 2λ . (d) Recovering the fold sequence and adding it back to the folded samples restores the original waveform. Panels (b)–(d) zoom on the shaded window of (a).

1.2 Problem Statement and Motivation

The dynamic range limitations of the US RF receive chain are managed in current scanners by placing a variable-gain amplifier before the converter and applying logarithmic compression to the displayed image. However, this fails to restore signals clipped by the converter because lowering the gain to protect a bright interface attenuates the faint echoes arriving simultaneously, while post-digitisation compression merely reshapes the surviving amplitudes. Alternative techniques acquire the same frame across multiple transmissions at different gains and merge the resulting records [3]. While this expands the captured range, it demands additional pulse transmissions per frame and relies on the assumption that the target anatomy remains stationary between acquisitions. A solution capable of expanding dynamic range within a single acquisition at the converter level is therefore of significant practical value.

The USF provides such a solution in principle, as its theoretical recovery guarantees apply to any bandlimited signal [2]. However, an US RF record consists of a sequence of short, high-frequency oscillatory wavelets. A bright acoustic reflector generates an echo that crosses the folding threshold of the modulo ADC repeatedly within a few cycles, producing a dense, rapidly alternating fold sequence. Furthermore, the physical receive chain introduces electronic noise prior to folding, quantises the modulo samples to a finite bit depth, and exhibits hardware wrap-height imperfections. Evaluating whether the fold sequence r can be reliably reconstructed from modulo measurements under these distortions, at a threshold low enough to deliver a meaningful dynamic range extension, forms the core focus of this investigation.

This work therefore determines the operating envelope within which an algorithm based on the USF successfully reconstructs US RF signals from modulo-folded samples. This envelope is characterised by the maximum allowable reduction of the folding threshold relative to the peak signal amplitude—representing the net dynamic range gain—and by the recovery algorithm’s tolerance to channel noise, coarse quantisation, and modulo wrap errors before reconstruction degrades.

1.3 Aim, Objectives, and Scope

1.3.1 Research Aim

The primary aim of this work is to determine, through simulation, the operating regime over which the implemented algorithm successfully recovers ultrasound RF signals from centred-modulo samples, thereby extending the achievable dynamic range without information loss from ADC saturation.

1.3.2 Research Objectives

To achieve this aim, the following specific objectives are addressed:

1. **Algorithm Development:** To design and implement a robust software recovery pipeline based on the USF that estimates the fold sequence of an ultrasound RF record and accurately reconstructs the original high-dynamic-range waveform from its modulo samples.
2. **Dynamic Range Evaluation:** To quantify the maximum extent to which the folding threshold λ can be lowered below the peak signal amplitude while maintaining exact signal reconstruction.
3. **Robustness and Envelope Mapping:** To systematically evaluate the recovery pipeline's tolerance to key physical receive-chain non-idealities, specifically establishing operational limits against additive channel noise, sample quantisation, and modulo wrap errors.

1.3.3 Scope and Limitations

The boundaries and operational constraints of this study are defined as follows:

Simulation Environment

1. All evaluations are conducted entirely in software using simulated ultrasound records. RF channel data are generated using Field II [4], [5] for a 5 MHz linear-array transducer at a fixed sampling frequency of 100 MHz.

2. Simulated tissue media encompass point-target reflectors, hyperechoic structures, anechoic cysts, and diffuse speckle.

Hardware and Receive Chain Modelling

1. Modulo folding is modelled as a centred-modulo operator with a known, ideal, fixed threshold λ .
2. Non-idealities are incorporated via: additive electronic channel noise, finite-bit quantisation of folded samples, and random variations in the modulo wrap height.
3. Physical hardware implementation, circuit design, and acquisition of *in vivo* clinical patient data lie outside the scope of this work.

Methodological Boundaries

1. The recovery framework relies on temporal oversampling; methods for sub-Nyquist or reduced-rate sampling are not investigated in this study.
2. Image reconstruction is restricted to standard delay-and-sum B-mode imaging.
3. Tissue attenuation and depth-dependent signal-to-noise ratio degradation and centre-frequency downshifting are not modelled.

1.4 Significance of the Study

For ultrasound imaging, this work demonstrates the viability of an analogue-to-digital front end whose dynamic range can sit well below the peak echo amplitude without losing faint or bright acoustic reflectors to ADC saturation.

The study also constitutes the implementation of the USF theory in a practical application. The published literature on the USF does not make assumptions about the signal beyond its band support, whereas an application to US RF implies a signal with structural properties of its own that may be exploited in the recovery algorithm.

Finally, by studying the operational regime against key physical hardware non-idealities, this study provides a preliminary contribution to guide the future design and implementation of modulo-based front ends in ultrasound receive chains.

1.5 Notation

Important symbols that recur from this chapter onward are:

Folding threshold (λ): The half-width of the modulo ADC dynamic range, constraining every folded sample $y[k]$ strictly to the interval $[-\lambda, \lambda)$.

Fold fraction (FF): The ratio expressing the folding threshold relative to the absolute peak amplitude of the signal under test, defined as $\lambda = \text{FF} \cdot \max |g|$. A smaller fold fraction corresponds to a narrower converter range.

Fold sequence (r): The piecewise-constant sequence comprising integer multiples of 2λ discarded during the folding process, satisfying $g[k] = y[k] + r[k]$. In the USF literature, r is also referred to as the residual or staircase sequence.

2

Literature Review

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2.1 The Unlimited Sensing Framework and Modulo Recovery

The USF, introduced by Bhandari et al. [2], establishes a theoretical paradigm for acquiring high-dynamic-range analogue signals without clipping. By capturing an input waveform $g(t)$ through a modulo ADC with folding threshold λ , the process records bounded measurements $y[k] = \mathcal{M}_\lambda(g(kT_s)) \in [-\lambda, \lambda)$. This maps the original signal to $g = y + r$, where $r \in 2\lambda\mathbb{Z}$ denotes the unobserved piecewise-constant residual sequence (or staircase).

While modulo folding preserves signal information, the central algorithmic challenge lies in inverting the non-linear operator $\mathcal{M}_\lambda$. Bhandari et al. proved that exact, deterministic reconstruction is guaranteed for bandlimited signals provided the sampling rate exceeds the Nyquist rate by a fixed factor independent of the signal amplitude, with the order N of the difference operator growing logarithmically with the ratio of the peak amplitude to the folding threshold. The

core strategy relies on applying finite-difference operators to eliminate the continuous waveform component, isolating discrete fold discontinuities, and resolving the resulting integration constants over the lattice $2\lambda\mathbb{Z}$. The first-order case $N = 1$ reduces the sampling condition to the requirement that consecutive samples differ by less than λ (which is the case on which the algorithm implemented in this work rests). The mathematical formulation and stability properties of this recovery mechanism are detailed below.

2.1.1 Published Methods for Modulo Recovery

Various algorithmic strategies have been proposed to reconstruct r from modulo measurements y . In their foundational work [2], Bhandari et al. establish that the modulo mapping $\mathcal{M}_\lambda$ and the N -th order finite difference operator Δ^N commute under sufficient oversampling, yielding:

$$\mathcal{M}_\lambda (\Delta^N y) [k] = (\Delta^N g) [k] \quad (2.1)$$

Consequently, the N -th order difference of the fold sequence can be computed directly from the observed modulo samples:

$$\Delta^N r[k] = \mathcal{M}_\lambda (\Delta^N y) [k] - \Delta^N y[k] \quad (2.2)$$

To reconstruct r from $\Delta^N r$, the difference operator is inverted via a running summation, constraining intermediate integrated sequences to the discrete lattice $2\lambda\mathbb{Z}$ while recursively resolving the unknown integration constants at each stage. This framework also provides theoretical stability guarantees under bounded deterministic noise, that is, perturbations whose amplitudes remain strictly below a known bound. For instance, rounding a modulo sample to B bits introduces quantisation noise, which acts as a bounded perturbation. The N -th order difference inflates that perturbation by up to 2^N . Due to a larger dynamic range demanding a larger N , the tolerable perturbation shrinks as the dynamic range grows. Robustness can be bought back by oversampling further, but recovery will be exact up to the perturbation itself, meaning the fold sequence is returned exactly, but the record is not denoised.

Subsequent research by Florescu et al. [6] bridges the gap between the mathematical formulation and the hardware implementation. In an ideal modulo encoder, when a signal crosses the positive or negative hardware threshold ($\pm\lambda$), it instantly resets to the opposite threshold ($\mp\lambda$), representing an exact voltage displacement of 2λ . In real analog circuits, there is almost always a mismatch or

misalignment between the trigger threshold and the post-reset voltage. Florescu et al. capture this mismatch using a hysteresis parameter, where instead of folding a full 2λ , the system resets to a set of asymmetric post-reset values defined by $\{\lambda-h, -\lambda+h\}$. This misalignment essentially means the encoder has "memory," where the output at any given moment depends on the historical trajectory of the signal. Hysteresis proves to be useful because it prevents folding events from occurring arbitrarily close in time, a common failure point in ideal modulo encoders where signals hovering near the threshold produce overlapping discontinuities that break reconstruction algorithms. The paper also introduces the concept of the folding transient, to characterize the non-zero transition period required for a physical analog circuit to complete a voltage reset. Florescu et al. model this transition as a continuous, piecewise-linear ramp. The foundational unlimited sensing algorithm fails in this regime because any discrete sample that lands directly on the transient ramp does not align with the ideal $2\lambda\mathbb{Z}$ grid; this introduces distortions that are recursively amplified during the integration step. To resolve this, the authors enforce the constraint that the sampling period exceed the transient duration, which mathematically guarantees that at most one discrete sample can land on the transient ramp during any single folding event. By convolving the non-ideal modulo samples with a higher-order finite-difference filter, the bandlimited signal component is suppressed, transforming the unobserved fold sequence into highly localized, high-amplitude pulse clusters centered around the discrete folding times. Leveraging the minimal temporal separation guaranteed by hysteresis, these clusters do not overlap under sufficient oversampling, allowing the algorithm to detect the asynchronous folding times and their directional signs. Once these folding parameters are resolved, the continuous residual function is reconstructed even from samples captured directly during the transient intervals.

In "Unlimited Sampling from Theory to Practice: Fourier-Prony Recovery and Prototype ADC" [7], the authors introduce an alternative approach designed to handle hardware non-idealities, such as delayed folding, spurious jumps, or uncertainties in the threshold λ . In practical settings, these imperfections mean that the folding residue $r[k]$ no longer lies strictly on a grid of $2\lambda\mathbb{Z}$. This causes the original commutative modulo-difference relation to fail, leading to significant reconstruction errors in the first algorithm [2]. To circumvent this, they propose a Fourier-domain recovery algorithm that is agnostic to the value of λ . By taking only the first-order finite difference of the modulo samples:

$$(\Delta y)[k] = (\Delta g)[k] - (\Delta r)[k]$$

the folding discontinuities manifest as sharp, sparse spikes $(\Delta r)[k]$. In the frequency domain, because the original signal is bandlimited, and the finite difference preserves that band, the spectrum

of $(\Delta g)[k]$ vanishes outside the passband. Outside the passband, the out-of-band Discrete Fourier Transform (DFT) coefficients of the observed differences $(\Delta y)[k]$ are therefore those of the folding spikes alone. This formulation transforms the reconstruction into a classical spectral estimation problem, as the out-of-band Fourier coefficients represent a sum of complex exponentials. Using high-resolution spectral estimation techniques, specifically Prony's method or the Matrix Pencil method, the algorithm estimates the exact locations and arbitrary real heights of the folding jumps directly from the out-of-band coefficients. Once the jump sequence $(\Delta r)[k]$ is resolved, it is integrated via a running summation to recover the original high-dynamic-range samples up to a single constant offset.

A more recent paper by Kvich et al. [8] treats the folding parameter λ as unknown and allows it to vary dynamically from sample to sample. The authors construct a uniqueness proof by framing modulo recovery as a "missing-sample" problem. Because any bandlimited signal with finite energy must decay to zero as time approaches infinity, its far-out tails are mathematically guaranteed to eventually drop below the threshold λ and remain completely unfolded. By temporarily setting aside the central folded region and treating those samples as "missing," it is proven that the continuous-time signal can be uniquely identified from these unfolded tails alone. The core of this uniqueness argument is that if two different bandlimited waveforms were to produce the same observed modulo data, they would have to be completely identical on their infinite tails; complex analysis dictates that two bandlimited signals sharing identical tails must be identical everywhere, proving that only a single, unique signal can fit the observed record even when the folding parameter is completely unknown.

This reconstruction, however, is highly unstable in the presence of noise. To construct a robust, practical recovery algorithm, Kvich et al. utilize the entire signal record by exploiting a frequency-separation constraint. Because the out-of-band frequency spectrum of the original bandlimited signal g is zero, any out-of-band components observed in the folded measurements must be a direct byproduct of the folding discontinuities. To solve for the folding events, the algorithm minimizes this out-of-band frequency residual. To make the optimization tractable, the integer constraint allows c to take real values, formulating a convex objective function.

$$\arg \min_{c \in \mathcal{L}_2} \|\mathcal{T}y + 2\hat{\lambda}\mathcal{T}c\|_2^2 + \mu\|Dc\|_1.$$

The solution is then rounded to the nearest integers, and the unknown folding threshold λ is subsequently re-estimated in closed form. This out-of-band residual-minimization framework achieves

robust recovery under additive white Gaussian noise across various oversampling factors, representing a significantly broader noise setting than prior algebraic unwrapping techniques.

A further development of the USF literature extends modulo recovery from lowpass to bandpass signals and purposefully aliases the bandpass signal's high-frequency spectrum into the vacant lowpass region through undersampling. A bandpass signal occupies a narrow band around a high carrier frequency, and this method exploits the empty spectrum below the band. Sampling slower than the Nyquist rate makes copies of the spectrum which can make them overlap, but if the rate is chosen so that the copies of the occupied band fall into the empty region without touching one another, the signal survives intact and can be recovered at a rate set by its bandwidth instead of its highest frequency. Shtendel et al. [9] carry this idea into the USF. Folding makes the signal non-bandlimited, so aliased copies can no longer avoid one another, yet the authors show that recovery survives, and their Unlimited Bandpass Sampling Theorem states the admissible sampling periods.

The permitted undersampling grows with the ratio of carrier frequency to bandwidth, so the regime rewards narrow fractional bandwidths. The guarantees they developed also carry over to imperfect folding hardware, including hysteresis and folds of arbitrary height.

2.1.2 Theoretical Boundaries and Assumptions of Existing USF literature

Viewed together, these methods can be categorised by the domain in which they identify the fold sequence. In the time-domain the methods [2], [6] locate each fold on the sample grid, assuming a known folding model, which is an exact $2\lambda\mathbb{Z}$ lattice in the first case and a calibrated hysteresis and folding transient description in the second. In the frequency-domain, the methods [7], [8] identify the folds from the out-of-band spectrum of the differenced samples and no longer need any assumption about the jump amplitude, so they tolerate arbitrary jump heights and an unknown or varying threshold λ . Nevertheless, the Fourier–Prony guarantees assume a known number of folds, and the residual-minimisation approach relaxes the integer lattice and must recover it by rounding. Every reviewed method assumes a bandlimited input (or bandpass support in the case of [9]), trades oversampling against the dynamic range sought, and in one form or another limits how densely folds may occur. None assumes any structure in the signal being reconstructed beyond its band support. This presents an opportunity for domain-specific applications of the USF, such as ultrasound imaging, to leverage distinct, highly structured waveform geometries to implement a recovery method.

2.2 Structure of the US RF signal

A standard model for ultrasound simulation is Jensen's [10] description of the received pulse-echo signal, built on an inhomogeneous wave equation for soft tissue. The received signal is a chain of three convolutions. The first forms a pulse-echo wavelet from the excitation and the transducer's two-way electromechanical impulse response. The second scatters that wavelet once from the tissue, which Jensen represents as small local deviations of density and sound speed. The third is a spatial convolution with the pulse-echo spatial impulse response, so the finite aperture of the transducer sets each echo's arrival time and adds a geometric spread on top of the wavelet's own width. One received signal is the response for a single transducer position, so sweeping that position across the region of interest in the body yields a set of scan lines, and combining them forms the B-mode picture. This wave model underlies the Field II simulator [4], [5] used in this work.

The characteristic look of clinical ultrasound RF is due to the way different structures in tissue scatter acoustic waves. Scattering is governed by the size of the structure in comparison to the wavelength of the incident ultrasound wave. Objects many times larger than the wavelength, such as organ boundaries, reflect specularly, and the scattered wavefront approximates a replica of the object's shape [1, Sec. 8.2.1]. Structures much smaller than the wavelength scatter diffusively, and each one contributes almost nothing on its own, but their interference creates a granular texture referred to as speckle. This interference pattern depends on the tissue and on the measuring system, through the width of the wavelet and the beam, the centre frequency, the bandwidth, and the focusing [1, Sec. 8.4.3]. A clinical US RF record is a superposition of the two regimes. Strong events from large structures stand far above a continuous weak background of interfering scatterers.

This superposition is difficult for an ADC to capture. A line through speckle alone fluctuates about a modest level and has no dominant peak. A line through a bright reflector carries one echo far stronger than the rest of the record.

Regardless of what causes the wave to scatter, the contribution to the record is a short two-way wavelet, fixed by the excitation pulse and by the transducer's electromechanical impulse response, applied once on transmit and once on receive. It consists of a few cycles at the centre frequency, scaled by the scatterer's strength and delayed by the two-way travel time. Additionally, the spatial impulse response, which describes how the extent of the transducer face spreads the arrival of that wavelet over a short interval (because different parts of the face lie at different distances from the

scatterer) lengthens the wavelet away from the focus, but every echo remains a brief oscillation near the centre frequency. This holds in the lossless medium simulated here. In tissue, attenuation grows with frequency, so the centre frequency of the echoes drifts downward with depth and the wavelet broadens.¹

In simulations, scattering objects of both kinds are commonly represented as ensembles of ideal point scatterers with assigned amplitudes, arranged in the shape of the desired tissue [1, Sec. 8.3], with amplitudes drawn from a Gaussian distribution where random speckle is wanted. Field II implements Jensen’s model on such ensembles and computes the received record for a specified transducer array and excitation [4], [5].

2.2.1 Ultrasound RF and the Dynamic-Range Problem

Clinical ultrasound RF records strongly and weakly reflecting tissue simultaneously. Szabo illustrates this with typical echo levels along one beam path through the heart [1, Sec. 10.8.2]. The interface between the posterior heart wall and the lung reflects almost perfectly, close to 0 dB relative to an ideal reflector. Interfaces between soft tissues, such as fat against muscle or blood against the vessel wall, return echoes near -55 dB. The backscatter from blood itself sits at -70 dB. A single line can therefore span roughly 70 dB between its strongest reflector and the weakest, and a receiver must cover at least that range to detect blood without saturating on the lung. Each front-end amplifier in a clinical system typically covers 55 to 60 dB, repeated on every one of the 64 or 128 receive channels [1, Sec. 10.8.2]. Whatever fits through the amplifier is then digitised by a converter whose range imposes the same kind of limitation. A higher bit budget of the converter lowers the floor beneath the weak echoes, but it does not increase the admissible input range, so the strongest echoes may be affected by clipping.

Attempts to widen the captured range exist. Degirmenci, Perrin and Howe transferred the multiple-exposure principle of high-dynamic-range photography to ultrasound, acquiring the same view several times at different gain settings and compositing the results [3]. The composite reaches a range no single acquisition can, but every frame now costs several transmissions, which taxes the frame rate and assumes the anatomy holds still between them.

Folding the record, using the methods described in Section 2.1, offers a promising alternative.

¹The recovery procedure of Chapter 4 takes its search window and its reserved harmonic count from the centre frequency and the pulse band, so under attenuation those settings would have to follow the local centre frequency along the record. Attenuation is not modelled in this work, and this dependence is noted as a limitation in Chapter 6.

3

Background Theory

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This chapter gathers the mathematical basis for the work presented, including the signal model, the relevant USF theory, and the metrics used to evaluate recovery.

3.1 The Ultrasound RF Forward Model

3.1.1 Characteristics of the pulse and the RF record

A simulated ultrasound RF record is the voltage trace produced by a modelled transducer after it has transmitted a short acoustic pulse into a simulated phantom, and has received the echoes that

structures along the path of the pulse scatter back towards the transducer face. To represent what occurs physically, the structures of the phantom are modelled as scatterers, each standing for a small local deviation of density or sound speed from the surrounding tissue. Each scatterer returns a copy of the transmitted wavelet, scaled by its own strength and delayed by the two-way travel time $2d/c$, where d is the depth of the scatterer and c is the speed of sound, and the full record is the superposition of all such copies. Each active element of the transducer aperture receives its own version of those echoes, differing from the others in arrival time, in amplitude, and in the way neighbouring echoes overlap, because the path from each scatterer to each element has a different length and direction. Each element therefore produces a record of its own, referred to in this thesis as a receive channel. The channel records are delayed after digitisation, so that echoes originating from a chosen point in the tissue align across the elements, and are then summed, so that the echoes from that point add constructively while echoes from elsewhere do not. The result is an "amplitude line", or "A-line".

Jensen [10] derived the superposition of echoes from an inhomogeneous wave equation in which soft tissue is represented as a homogeneous medium carrying small local perturbations of density and propagation velocity. Under the assumptions of linear propagation and weak scattering (the first-order Born approximation), the received pressure signal for one transducer position is

$$g(t) = v_{\text{pe}}(t) \underset{t}{*} f_m(\mathbf{r}) \underset{\mathbf{r}}{*} h_{\text{pe}}(\mathbf{r}, t), \quad (3.1)$$

where $\underset{t}{*}$ denotes convolution in time and $\underset{\mathbf{r}}{*}$ denotes convolution over the scattering volume [10, Eq. 45]. The pulse-echo wavelet $v_{\text{pe}}(t)$ combines the transmitted excitation with the electromechanical impulse response of the transducer on transmission and on reception, and it fixes the shape of every individual echo. The scattering term $f_m(\mathbf{r})$ collects the density and velocity perturbations at each point of the tissue, and it is the quantity that an ultrasound image ultimately seeks to display. The pulse-echo spatial impulse response $h_{\text{pe}}(\mathbf{r}, t)$ accounts for the finite aperture of the transducer, and it determines how an echo from a given point is spread in time and how strongly it is received as a function of position relative to the beam. The record g is the transducer's electrical output we need. The program used Field II [4], [5] computes this for a specified transducer geometry, excitation, and scatterer distribution.

It is important to emphasize again that, based on (3.1), each echo is a copy of v_{pe} , which, for a transducer of centre frequency f_c , driven by an excitation of a few cycles, is a short oscillation at approximately f_c lasting a few periods, lengthened somewhat by h_{pe} away from the focus. Fig-

ure 3.1 shows this wavelet for the transducer simulated in Chapter 5, where a two-cycle excitation at $f_c = 5$ MHz passes through the electromechanical impulse response of the transducer once on transmission and once again on reception. An isolated strong reflector, such as a tissue boundary or a point target, contributes a single such wavelet of large amplitude at its own arrival time. A region of diffuse scatterers contributes a great many weak wavelets whose arrival times differ so that they interfere to form the granular speckle texture characteristic of soft tissue. The record formed is therefore a train of brief oscillations, of which a few may stand far above the rest.

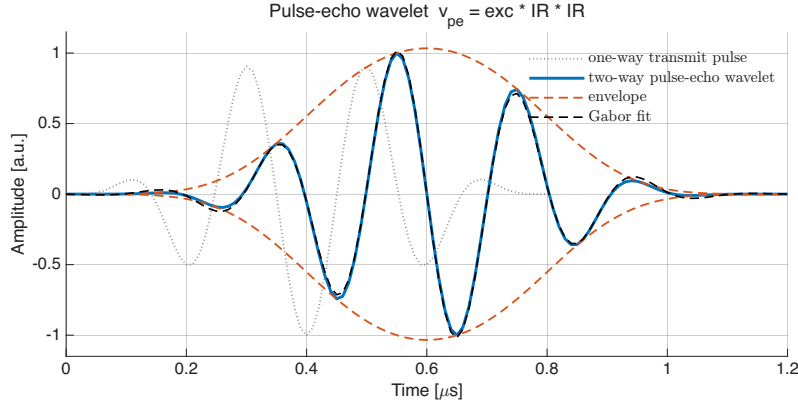


Figure 3.1: Pulse-echo wavelet v_{pe} of the transducer used in Chapter 5, formed as the excitation convolved with the transducer impulse response on transmission and again on reception. The dotted grey trace is the one-way transmitted pulse, a two-cycle excitation at $f_c = 5$ MHz passed once through the impulse response. The solid blue trace is the two-way wavelet, which is the shape that every echo in the record copies, and the dashed orange curves are its envelope. The dashed black trace is a Gabor function, a Gaussian-windowed cosine, fitted to the wavelet by least squares with a relative residual of 4%. The takeaway is that each echo is a windowed oscillation at f_c .

As stated before, the simulated records used in Chapter 5 ignore any effects of attenuation in the tissue.

3.1.2 Noise

Two sources of noise were modelled. The first is the electronic noise of the receive channel, which arises from thermal noise in the transducer element and the front-end amplifier, and it is uncorrelated from one element to the next, so it is modelled in the ultrasound literature as additive white Gaussian noise added independently to each channel record [11]. The noisy record is

$$g_n[k] = g[k] + \sigma \eta[k], \quad \eta[k] \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1), \quad (3.2)$$

where k indexes the samples taken at the sampling period T_s . The channel signal-to-noise ratio is defined against the root-mean-square level of the clean record ($\text{rms}(g)$):

$$\text{SNR} = 20 \log_{10} \frac{\text{rms}(g)}{\sigma}, \quad \text{rms}(g) = \sqrt{\frac{1}{K} \sum_{k=0}^{K-1} g[k]^2}, \quad (3.3)$$

so that a stated Signal-to-Noise Ratio (SNR) fixes σ through $\sigma = 10^{-\text{SNR}/20} \text{rms}(g)$.

The second source is the finite resolution of the converter, which acts after folding on samples that already lie within $[-\lambda, \lambda)$. A converter of B bits divides that range into 2^B equal steps of width $2\lambda/2^B$ and reports each sample rounded to the nearest step, so the recorded value differs from the folded value by at most half a step, a bounded error of $\lambda 2^{-B}$ in magnitude [2]. Physically this represents the quantisation of any digital converter.

A third hardware non-ideality, in which a folding circuit wraps by a height other than exactly 2λ , is defined in Section 3.2.

3.2 Modulo Sampling and the Fold Sequence

3.2.1 The Centred Modulo Map

The USF [2] implements a centred modulo map of the form,

$$\mathcal{M}_\lambda(g) = ((g + \lambda) \bmod 2\lambda) - \lambda, \quad (3.4)$$

which returns a value in $[-\lambda, \lambda)$. Figure 3.2 contrasts the two transfer curves, and Fig. 3.3 shows their effect on a window of ultrasound RF. The folded record is the sequence $y[k] = \mathcal{M}_\lambda(g[k])$.

3.2.2 The Fold Sequence

The map only ever subtracts multiples of 2λ , so the original record can be written exactly as the folded record plus what was subtracted [2],

$$g[k] = y[k] + r[k], \quad r[k] \in 2\lambda\mathbb{Z}. \quad (3.5)$$

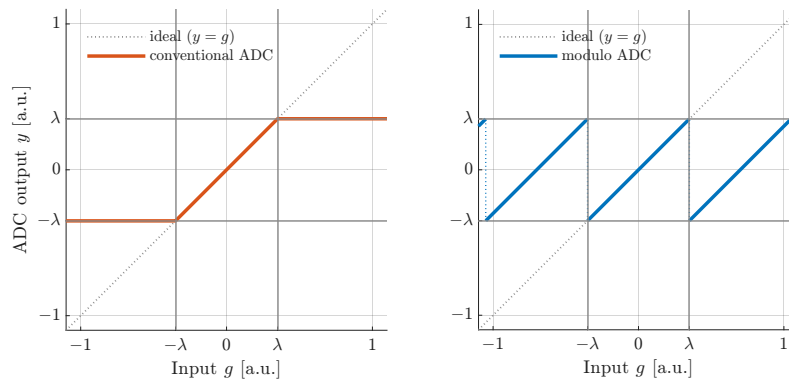


Figure 3.2: Input-output curves of the two converters at the same input range $\pm\lambda$. Left, a conventional converter. Right, a modulo converter. The dotted diagonal is the ideal converter $y = g$.

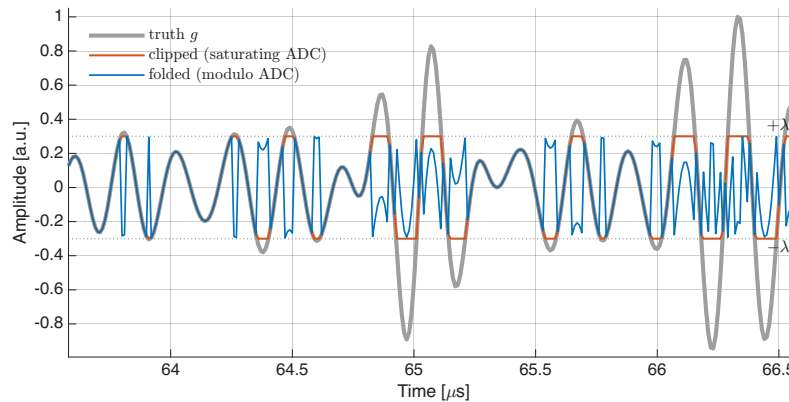


Figure 3.3: A window of a simulated speckle RF line with the converter rails set at 30% of the peak of the record. The clipped record replaces every excursion beyond $\pm\lambda$ by the rail value. The folded record wraps each excursion back into $[-\lambda, \lambda]$, so the waveform continues inside the window but with jumps of 2λ at every crossing of a rail $\pm\lambda$.

The sequence r is piecewise constant. It holds one value until the input crosses a rail, changes by exactly $\pm 2\lambda$ at that crossing, and holds the new value until the next. This is the "fold sequence" and its recovery is the whole problem, because once r is known the original record follows.

Writing the change of r between consecutive samples in units of 2λ ,

$$(\Delta r)[k] = r[k+1] - r[k] = 2\lambda q[k], \quad q[k] \in \mathbb{Z}, \quad (3.6)$$

gives the integer jump sequence $\mathbf{q} = q[k]$, which is zero wherever the input stayed inside the window, takes the value ± 1 where the modulo map subtracted or added one multiple of 2λ between two samples, and takes a larger magnitude only if the input moved by more than 2λ within a single sampling interval, so that several multiples were removed at once. The fold count M is the total number of unit jumps,

$$M = \|\mathbf{q}\|_1. \quad (3.7)$$

The jump sequence $\mathbf{q}$ records the position and direction of every fold, so recovering $\mathbf{q}$ is equivalent to recovering r , up to the value of r at the first sample.

The threshold λ is set relative to the peak of the record under test through the fold fraction FF:

$$\lambda = \text{FF} \cdot \max_k |g[k]|. \quad (3.8)$$

A smaller fraction describes a converter whose range covers a smaller part of the signal and therefore produces more folds.

3.2.3 Fold Pairs in Pulse-Echo Records

The pulse-echo structure of Section 3.1.1 gives the fold sequence of an ultrasound record a particular shape. Every echo is a brief oscillation of the form shown in Fig. 3.1. A crossing of $+\lambda$ on the way up is therefore followed, within a few samples, by a crossing of $+\lambda$ on the way down, and the corresponding entries of $\mathbf{q}$ are a $+1$ closely followed by a -1 , or vice versa at the lower rail. The folds of an RF record thus arrive as "closed pairs" of opposite sign (Figure 3.4 illustrates this structure). Chapter 4 leverages this structure to recover the fold sequence.

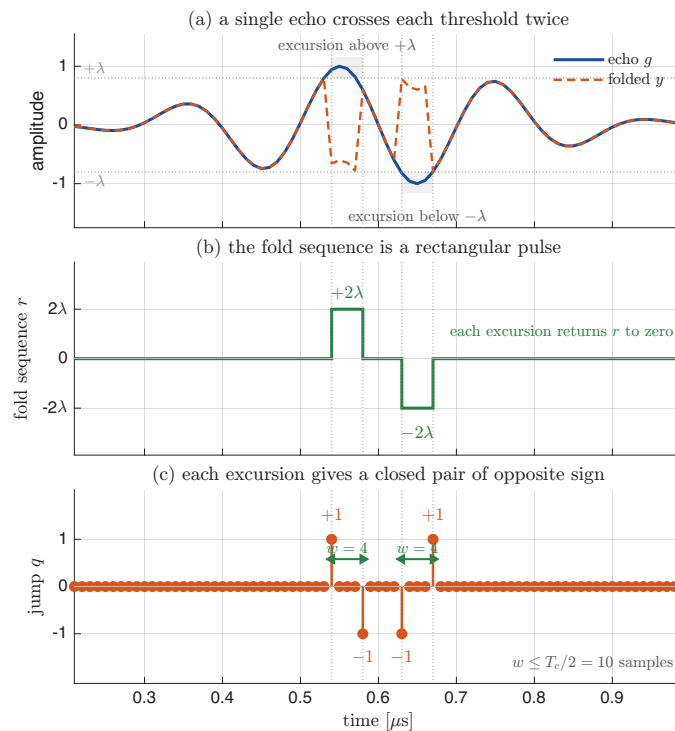


Figure 3.4: Illustration of what "closed-pairs" means, on an ultrasound record. (a) The echo crosses $+\lambda$ on the upward slope and returns across it a few samples later, repeating this behavior at $-\lambda$; consequently, the folded record y deviates from g only during these short excursions. (b) The resulting fold sequence r forms a rectangular pulse of height 2λ that returns to zero after each excursion. (c) In the jump sequence $\mathbf{q}$, each excursion manifests as a $+1$ closely followed by a -1 at the upper threshold, and in the reverse order at the lower threshold. Here, both pairs span $w = 4$ samples, well within the half-period bound $T_c/2 = 10$ samples at $f_c = 5$ MHz and $f_s = 100$ MHz, which Chapter 4 uses to set $w_{\max}$.

3.2.4 Fold-Amplitude Jitter

A physical folding circuit subtracts a fixed voltage each time the input crosses a rail, and that voltage may differ slightly from the intended value, so a wrap of $2\lambda(q \pm \varepsilon)$ occurs in place of $2\lambda q$, with ε a small fraction. Referred to as the "severity" of the jitter. Because the circuit holds a running offset, a wrap of the wrong height leaves that offset in error for every later sample until the next wrap corrects or compounds it. The integer sequence $\mathbf{q}$ is unaffected, since the fold still happened at the same sample in the same direction, but the fold sequence r is no longer an exact multiple of 2λ . Chapter 5 specifies how the jitter is drawn and Chapter 6 measures its effect.

3.3 Published Recovery Methods

3.3.1 Finite differences and the direct read

Let Δ denote the first-difference operator on a sequence, $(\Delta x)[k] = x[k+1] - x[k]$, for $k = 0, \dots, K-2$. Applying it to both sides of (3.5) and using (3.6) gives

$$\Delta \mathbf{g} = \Delta \mathbf{y} + 2\lambda \mathbf{q}, \quad \mathbf{q} \in \mathbb{Z}^{K-1}, \quad (3.9)$$

that is, the difference of the original record equals the difference of the folded record plus the jump sequence scaled by 2λ . Differencing turns the piecewise-constant fold sequence into a sparse sequence of impulses [2], one at each fold, and the positions and signs of those impulses are what recovery has to find. The fold sequence is rebuilt from its jumps by cumulative summation,

$$r[k] = r[0] + 2\lambda \sum_{j=0}^{k-1} q[j], \quad (3.10)$$

so that $\mathbf{q}$ determines r up to the value at the first sample. This work assumes that every record begins inside the converter window, and therefore takes $r[0] = 0$.

The jumps $\mathbf{q}$ can be read directly from the folded record when the original signal moves slowly enough between samples. If consecutive samples of g never differ by as much as the threshold λ ,

$$\|\Delta \mathbf{g}\|_{\infty} < \lambda \implies \mathbf{q} \in \{-1, 0, +1\}^{K-1}, \quad (3.11)$$

then no gap between samples can contain more than one fold, and the folded difference $\Delta \mathbf{y}$ can

differ from $\Delta \mathbf{g}$ only by a single multiple of 2λ . We abide in this work by condition (3.11), Also referred to as Itoh's condition [12], to build the recovery algorithm [2].

A large difference in the folded record then naturally corresponds to the trace of a fold, and its sign gives the direction of the wrap, so that

$$\hat{q}[k] = \text{round}\left(-\frac{(\Delta y)[k]}{2\lambda}\right) \quad (3.12)$$

recovers $\mathbf{q}$ exactly and, through (3.10), recovers the fold sequence.

The condition (3.11) is a property of the signal and of the sampling rate relative to the threshold. Where it holds, (3.12) can be read off the data directly and needs no further computation. Where it fails, the read is wrong in that gap, and every later value of (3.10) carries that error. Chapter 4 uses (3.12) as the starting estimate of the recovery algorithm.

3.3.2 Out-of-band coefficients and the scoring of a candidate fold sequence

To verify whether the direct read (3.12) gives a correct estimate of the jump sequence, we turn to the frequency domain. The record g is bandlimited, thus, its difference $\Delta \mathbf{g}$ occupies only a limited set of bins of the DFT. Let P , the reserved harmonic count, be the highest bin index that the spectrum of the pulse reaches, so that the in-band bins are those within P of either end of the spectrum and the remaining L bins are out of band. Folds are jump discontinuities and therefore broadband, so they populate L exclusively. Writing $\mathbf{V}$ for the DFT matrix of length $K - 1$ and $\mathbf{S}$ for the matrix that selects the L out-of-band rows, the difference relation (3.9) gives

$$\mathbf{z} = \mathbf{SV} \Delta \mathbf{y} = -2\lambda \mathbf{SV} \mathbf{q} + \mathbf{w}, \quad (3.13)$$

$\mathbf{z}$ is the vector of DFT out-of-band coefficients, $\mathbf{w}$ corresponds to channel noise and any leakage of the pulse spectrum past P . Chapter 4 describes how P is chosen in this work.

$\mathbf{z}$ serves to score any proposed jump sequence. A candidate $\mathbf{q}'$ predicts, through (3.13), the out-of-band content $-2\lambda \mathbf{SV} \mathbf{q}'$. We can measure how well the candidate matches the true out-of-band content by taking the norm of the difference:

$$\rho(\mathbf{q}') = \|\mathbf{z} + 2\lambda \mathbf{SV} \mathbf{q}'\|_2, \quad (3.14)$$

$\rho(\mathbf{q}')$ is small when the candidate accurately reproduces the true folds and large when it does not. When $\mathbf{q}' = \mathbf{q}$, the expression (3.14) reduces to $\|\mathbf{w}\|_2$, the noise alone. Kvich et al. recover the fold sequence by minimising this out-of-band misfit over real-valued jumps and rounding the result [8]. A similar idea is implemented in Chapter 4, where ρ is used first to test whether the reading obtained by (3.12), $\hat{\mathbf{q}}$, explains the out-of-band content and, if it does not, to rank alternative jump sequences, and to keep the one with the smallest misfit.

The alternative jump sequences are generated by two search strategies: "Matching Pursuit" [13] builds a sparse representation one element at a time, at each step adding the element of a fixed dictionary that reduces the residual the most. "Beam search" [14, Sec. 13.4] extends this greedy algorithm by retaining several partial solutions at once. In the form described by Jurafsky and Martin, every retained partial solution is extended by every element of the dictionary, all of the extensions are scored, and only the W best survive to the next level, so that an early choice which looks poor on its own can still lead to the best complete solution. In the implementation of Chapter 4 the dictionary consists of pairs of opposite unit jumps, and the score is the misfit ρ . Whereas Matching Pursuit scales each selected element by a fitted real amplitude, here every pair enters with its two jumps fixed at exactly $\pm 2\lambda$, so that each candidate remains a fold sequence that the converter could have produced. The search runs to the fixed depth of $\widetilde{M}/2$ pairs (where $\widetilde{M}$ is the estimated fold count), and the winning train is "refined" by exchanging one pair at a time for any pair of the dictionary that lowers the misfit.

3.4 Delay-and-Sum Beamforming and B-mode Display

3.4.1 Delay-and-sum beamforming

A scan line is formed from the channel records by delay-and-sum. Let the line pass through the centre of the active aperture and let the focal point lie on that line at depth z_r . An echo from the focus reaches the element at lateral offset x_n from the line after a path that is longer than the path to the aperture centre, so each channel is shifted by

$$\tau_n = \frac{z_r - \sqrt{x_n^2 + z_r^2}}{c}, \quad (3.15)$$

which aligns the arrivals from the focus across the aperture [1, Sec. 7.4.6] (c is the speed of sound in the medium). The same delays serve on transmission, where they are applied before firing, and

on reception, where they are applied to the recorded channels. The aligned channels are weighted by an apodisation window that tapers the edges of the aperture and are then summed, so that echoes from the focus add coherently while echoes from elsewhere largely cancel. The experiments of Chapter 5 use a translating aperture of 64 elements with Hamming apodisation and one fixed focus per line.

3.4.2 Envelope detection and log compression

The oscillation at the centre frequency carries no diagnostic information of its own, so the beam-formed line is reduced to its envelope, obtained as the magnitude of the analytic signal formed with the Hilbert transform, and the envelope is then compressed logarithmically to a display scale in decibels. The resulting grey levels, arranged line by line, form the B-mode image [1, Sec. 8.4.1]. The differences in pixel intensity reported in Chapter 6 are measured on the log-compressed image. Every image is compressed against the same reference level (the peak envelope of the ideal image formed without a converter), such that a difference in pixel intensity reflects a difference in recovered amplitude.

3.5 Error Metrics

At signal level, recovery of the fold sequence is called exact when the estimated jump sequence $\hat{\mathbf{q}}$ equals the reference jump sequence $\mathbf{q}$ in every entry, that is, in the position and sign of every fold. The amplitude error of the unfolded record is reported as a normalised root-mean-square error,

$$\text{NRMSE} = 100\% \times \frac{\|(\hat{\mathbf{g}} - \bar{\hat{\mathbf{g}}}) - (\mathbf{g}_n - \bar{\mathbf{g}}_n)\|_2}{\|\mathbf{g}_n - \bar{\mathbf{g}}_n\|_2}, \quad (3.16)$$

where $\hat{\mathbf{g}} = \mathbf{y} + \hat{\mathbf{r}}$ is the unfolded record, $\mathbf{g}_n$ is the noisy record of (3.2) that was folded, $\|\cdot\|_2$ is the Euclidean norm, and a bar denotes the mean of a record over all its samples. Subtracting each record's mean before comparing them makes the error insensitive to a constant offset between $\hat{\mathbf{g}}$ and $\mathbf{g}_n$, which is appropriate because the cumulative sum in (3.10) determines the fold sequence only up to its starting value. When fold-amplitude jitter is present, an exact jump sequence obviously does not imply a small Normalised Root-Mean-Square Error (NRMSE), because the wrapping height in the record is no longer an exact multiple of 2λ . Chapter 5 specifies the reference used in that case.

At the image level, the B-mode image formed after each channel is recovered, is compared pixel by pixel with two references formed from the same noisy channels, one ideal and one with the saturating converter of the same input range $\pm\lambda$. The reported quantities are the maximum and the mean of the absolute pixel difference in decibels on the log-compressed image. Contrast ratio, generalised contrast-to-noise ratio, and speckle signal-to-noise ratio are common measures of ultrasound image quality, but they require regions of interest that the experiments do not define, and they are not reported.

Fold-Sequence Recovery for Ultrasound RF

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4.1 Problem Statement

Given K centred-modulo samples y of an ultrasound RF record acquired at a known folding threshold λ , the primary objective is to estimate the piecewise-constant fold sequence r and reconstruct the original signal g across its full dynamic range.

Recall from the centered-modulo model (Equations (3.4) and (3.5)) that the ground-truth waveform decomposes as $g = y + r$, where $r \in 2\lambda\mathbb{Z}$ represents the step-wise residual discarded by the folding operation, $r = g - \mathcal{M}_\lambda(g)$. Consequently, recovery reduces to identifying the discrete jump locations and integer magnitudes that define r . Adding the estimated fold sequence $\hat{r}$ back to the observed modulo measurements restores the original amplitude information, yielding $\hat{g} = y + \hat{r}$. Figure 4.1 illustrates this decomposition for a representative RF A-line.

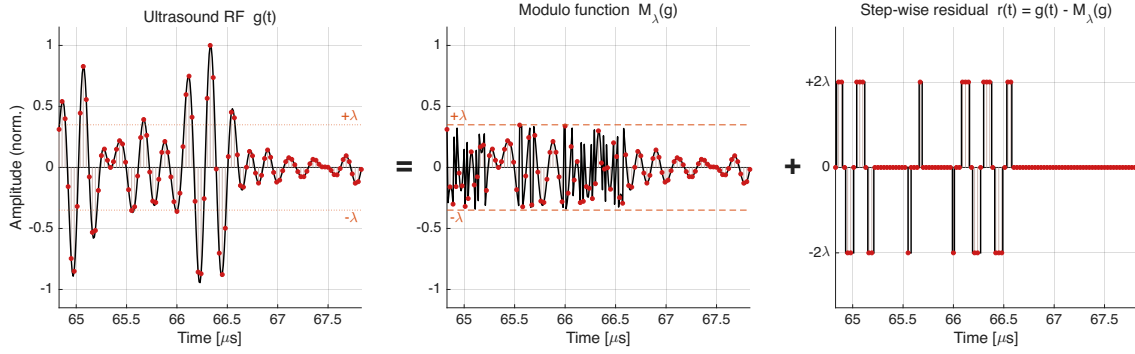


Figure 4.1: Centered-modulo decomposition of an RF A-line. Left: Ground-truth RF signal $g(t)$ relative to folding thresholds $\pm\lambda$. Middle: Centered-modulo signal $\mathcal{M}_\lambda(g)$. Right: Step-wise residual fold sequence $r(t) = g(t) - \mathcal{M}_\lambda(g)$. Unfolding restores $g(t)$ by adding $\hat{r}(t)$ back to the modulo samples.

4.2 Baseline for the Work Presented

A first attempt at recovering US RF from centred-modulo samples followed the algorithm presented in Bhandari, Krahmer and Poskitt [7]. After first differencing, a DFT is taken of the differenced modulo record. As discussed previously, the *out-of-band bins* are the frequency indices outside the RF bandlimit, where a bandlimited record contributes (ideally) no energy, so these bins contain only the fold residual and noise. Let $\mathbf{z}$ denote the vector of Fourier coefficients on those bins. Figure 4.2 shows how a reserved harmonic count P marks that bandlimit on an RF line, and how the folds fill the previously empty bins that make up $\mathbf{z}$.

Each fold becomes an impulse in the difference domain, leaving a characteristic trace on the out-of-band coefficients, which is a pure complex exponential whose frequency encodes where the fold occurred. With M folds, $\mathbf{z}$ is a mixture of M tones, and recovering the fold locations can be accomplished through spectral estimation. Bhandari et al. [7] solve this with an annihilating filter in Prony form, a short filter constructed from a Toeplitz system in $\mathbf{z}$ whose zeros are the tone frequencies. For the spectral step under noise the same authors point to the Matrix Pencil method [15]. The baseline implemented for this comparison uses its Singular Value Decomposition (SVD) form [16], which replaces the filter solve with a Hankel embedding of $\mathbf{z}$.

In the Matrix Pencil method, overlapping snippets of $\mathbf{z}$ are stacked into a Hankel matrix, an SVD extracts an M -dimensional signal subspace, and that subspace is compared with a one-sample-shifted copy of itself [16]. Shifting a tone ξ^k multiplies it by ξ , so the eigenvalues of the unshifted-to-shifted map are the poles ξ_m . Each pole's phase gives a fold location. A least-squares

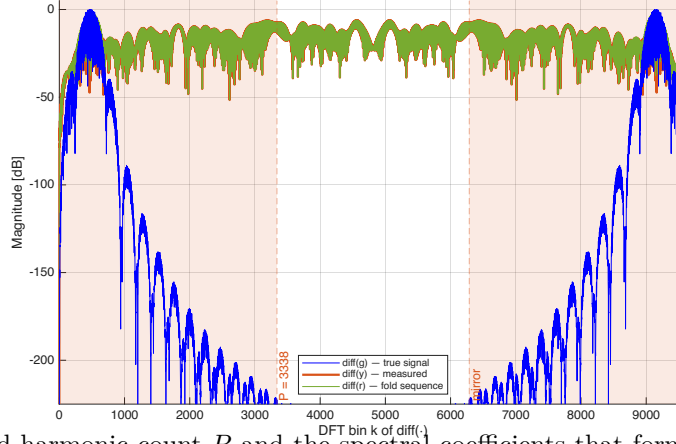


Figure 4.2: Reserved harmonic count P and the spectral coefficients that form $\mathbf{z}$, on the example RF line 17 after centred-modulo folding at $FF = 0.32$ ($M = 32$ folds), the same record and threshold as Fig. 4.5. Orange shading marks the reserved bands $[0, P]$ and $[K-1-P, K-1]$, with $P = 3338$ and $K-1-P = 6285$ ($f_P \approx 34.7$ MHz), and the white middle is where $\mathbf{z}$ is taken. The blue trace is the unfolded difference Δg , which leaves the middle empty. The orange trace is the folded difference Δy , and the green trace is the difference of the fold sequence, $\Delta r = 2\lambda \mathbf{q}$. The two coincide across the white middle, so the out-of-band coefficients of the folded record are the spectrum of the fold sequence alone.

fit then estimates each jump amplitude, a cumulative sum rebuilds the fold sequence, and adding it to y unfolds the record up to an additive constant, following the reconstruction steps of [7].

In US RF, bright echoes create folds in tight opposite-sign “pairs”. The signal crosses $+\lambda$ on the way up and re-crosses a few samples later on the way down, leaving a $+2\lambda$ jump and a -2λ jump almost on top of each other. The two tones sit at nearly identical frequencies with opposite-sign amplitudes, so their out-of-band signatures nearly cancel. The evidence that two distinct folds exist at all is carried by components several orders of magnitude weaker than the dominant ones, and even mild noise buries that information. Estimated locations then miss by one or two samples, the amplitude fit degrades with them, and unfolding integrates every wrong amplitude into a lasting offset. The SVD truncation used inside the Matrix Pencil improves numerical conditioning of the spectral step [16], but it does not remove the problem on the RF records studied here. Figure 4.3 shows this failure on an RF record used throughout Chapters 5 and 6, at a fold fraction of $FF = 0.90$, producing only four folds, and at a channel SNR of 80 dB.

In summary, the difficulty lies in how the folds are estimated. The out-of-band coefficients still carry the full signature of every fold, as Fig. 4.2 shows, but the attempt to read the fold positions as the poles of a mixture of tones fails, when two of those tones nearly cancel. The remainder of this chapter therefore abandons parametric estimation. Instead of estimating the jump amplitudes, the next method only considers the known values $\pm 2\lambda$ that the converter imposes. Instead of

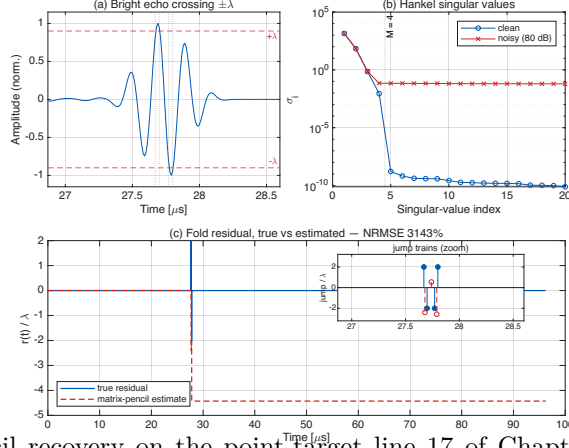


Figure 4.3: Matrix Pencil recovery on the point-target line 17 of Chapter 5 at $FF = 0.90$ and 80 dB channel SNR, with the true fold count $M = 4$ supplied to the estimator. (a) The brightest echo crosses the rails $\pm\lambda$ twice, so the record holds two closed fold pairs, whose fold instants are marked by the dotted lines. (b) Singular values of the Hankel matrix built from $\mathbf{z}$, for the noiseless record and for the same record with channel noise. Without noise the four fold tones stand clear of the numerical floor, but the fourth singular value lies several orders of magnitude below the first two, and the noise at 80 dB raises the floor above it, so only three of the four tones survive the subspace step. (c) The true fold sequence and the Matrix Pencil estimate over the whole record, with the four jumps and their estimates enlarged in the inset. The estimated jumps land at the wrong heights, and the cumulative sum carries that error as a constant offset of several λ over the remainder of the record.

estimating positions from $\mathbf{z}$, it reads them directly from the differenced samples and restricts them to the closely spaced pairs of opposite sign that the shape of the US pulse dictates. And uses $\mathbf{z}$ exclusively to check whether a candidate fold sequence explains the out-of-band content. The procedure, built on the difference identity of [2] and the out-of-band construction of [7], is set out in the next section.

4.3 Proposed Recovery Procedure

N.B. Some equations from previous chapters are repeated here for clarity.

The first-difference operator Δ turns r into a sequence of jumps $(\Delta r)[k] = 2\lambda q[k]$ (see Bottom-Right of Fig. 4.4). Applied to (3.5), the first difference gives

$$\Delta \mathbf{g} = \Delta \mathbf{y} + 2\lambda \mathbf{q}, \quad \mathbf{q} \in \mathbb{Z}^{K-1}, \quad M = \|\mathbf{q}\|_1, \quad (4.1)$$

where the support of $\mathbf{q}$ marks the folding instants, M is the number of folds, and the jump at index k has amplitude $2\lambda q[k]$. Recovering the integer jump sequence $\mathbf{q}$ thus recovers the residual by

cumulative summation, up to the unknown starting value $r[0]$, and solves the unfolding problem [2]. Throughout this work the record is assumed to begin unfolded, so $r[0] = 0$.

Under Itoh's condition (3.11), each nonzero entry is one unit fold, so amplitudes need not be estimated continuously [2], [12]. Figure 4.4 offers a visual walkthrough of (3.5) and (4.1) on one RF excerpt .

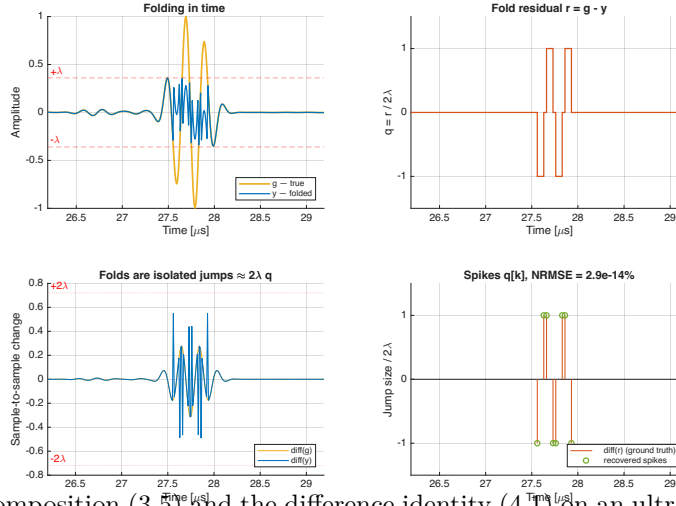


Figure 4.4: The decomposition (3.5) and the difference identity (4.1) on an ultrasound RF excerpt. Top-Left, g and its centred-modulo version y at rails $\pm\lambda$. Top-Right, residual $r = g - y$ on the 2λ lattice. Bottom-Left, Δg and Δy , which isolate folds as sparse jumps. Bottom-Right, unit spikes $\mathbf{q} = \Delta r / (2\lambda)$ and the recovered $\hat{\mathbf{q}}$, coincident on this clean excerpt.

The remaining task is to estimate $\mathbf{q} \in \{-1, 0, +1\}^{K-1}$. Two restrictions make the search over such sequences finite. First, folds occur around local cues in y , that is, when a sample approaches the rails, $|y[k]| > \eta_g \lambda$, or when the recorded step across a gap is abruptly large, $|(\Delta y)[k]| > \eta_j \lambda$, where η_g and η_j are fixed cue thresholds (values in Table 4.1). The marked gaps, widened by $\pm d$ samples to absorb small timing uncertainty, form an index set $\mathcal{G}$:

$$\mathcal{G} = \left(\{k : |y[k]| > \eta_g \lambda\} \cup \{k : |(\Delta y)[k]| > \eta_j \lambda\} \right) \oplus \{-d, \dots, d\}. \quad (4.2)$$

Second, the recorded RF is built from short pulse-echo wavelets, so a bright echo that leaves $[-\lambda, \lambda)$ returns within a small number of samples, never more than a fixed width $w_{\max}$ (value in Table 4.1). The corresponding jumps in $\mathbf{q}$ then form a closed fold pair: a $+1$ exit at some gap p linked to a -1 re-entry at $p + w$ with $1 \leq w \leq w_{\max}$, or the opposite orientation. Collect every admissible pair into a catalogue

$$\mathcal{C} = \{(p, p+w) : p, p+w \in \mathcal{G}, 1 \leq w \leq w_{\max}\}. \quad (4.3)$$

Writing $\mathbf{e}_p$ for the sequence that is one at gap p and zero elsewhere, the candidate fold sequences are the signed combinations of $M/2$ such pairs,

$$\mathcal{Q} = \left\{ \mathbf{q}' = \sum_{i=1}^{M/2} s_i (\mathbf{e}_{p_i} - \mathbf{e}_{p_i+w_i}) : s_i \in \{\pm 1\}, (p_i, p_i+w_i) \in \mathcal{C} \right\}, \quad (4.4)$$

a finite set in which every candidate fold is paired up with a nearby candidate fold of opposite sign, such that $\sum_k q'[k] = 0$.

The out-of-band coefficient vector $\mathbf{z}$ is used to score candidates. With $\mathbf{V}$ the DFT matrix of a length- $(K-1)$ sequence and $\mathbf{S}$ selecting the L out-of-band rows, the bandlimited part of $\Delta \mathbf{g}$ contributes nothing to those bins, so the measurement is

$$\mathbf{z} = -2\lambda \mathbf{S} \mathbf{V} \mathbf{q} + \mathbf{w}, \quad (4.5)$$

where $\mathbf{w}$ collects noise. The signature of a unit jump at gap k is the k -th column of $\mathbf{S} \mathbf{V}$, obtained by applying it to the unit impulse $\mathbf{e}_k$,

$$\mathbf{v}_k = \mathbf{S} \mathbf{V} \mathbf{e}_k, \quad (4.6)$$

A candidate $\mathbf{q}'$ then predicts the signature $2\lambda \mathbf{S} \mathbf{V} \mathbf{q}'$. Its misfit ρ ,

$$\rho(\mathbf{q}') = \|\mathbf{z} + 2\lambda \mathbf{S} \mathbf{V} \mathbf{q}'\|_2 \quad (4.7)$$

is small when the jumps are right and large when they are wrong [7] [8]; at the true fold sequence it reduces to the noise alone, $\rho(\mathbf{q}) = \|\mathbf{w}\|_2$.

Wherever (3.11) holds, it is possible to do a direct read of the fold vector from the data, as well as obtain the number of folds [2], via

$$\hat{q}[k] = \text{round}\left(-\frac{(\Delta y)[k]}{2\lambda}\right), \quad \widetilde{M} = \|\hat{\mathbf{q}}\|_1. \quad (4.8)$$

The *Read* panel of Fig. 4.5 shows the result of the read (4.8) on an ultrasound echo. The result is used as the seed hypothesis for verification against $\mathbf{z}$ and, if needed, for later refinement. The seed is accepted as is when it fits inside $\mathcal{Q}$ and explains $\mathbf{z}$ well enough, that is, when its misfit satisfies $\rho \leq \theta \|\mathbf{z}\|_2$ for a small fixed fraction θ (value in Table 4.1), or when a check on the leftover shows that any excess misfit is consistent with noise rather than a missing jump¹ (see the *Verify* panel

¹This verification uses a matched-filter on the spectral residual vector $\mathbf{u} = \mathbf{z} + 2\lambda \mathbf{S} \mathbf{V} \hat{\mathbf{q}}$. Substituting forward model (4.5) yields $\mathbf{u} = 2\lambda \mathbf{S} \mathbf{V} (\hat{\mathbf{q}} - \mathbf{q}) + \mathbf{w}$, isolating the read error. A single unmodelled fold at gap index k corresponds

of Fig. 4.5). If those conditions fail, the method falls back on the minimisation

$$\hat{\mathbf{q}} = \arg \min_{\mathbf{q}' \in \mathcal{Q}} \rho(\mathbf{q}'), \quad (4.9)$$

which cannot be solved exactly, because the number of candidates in $\mathcal{Q}$ grows combinatorially with the number of folds. It is approximated in two stages. The first stage keeps the seed and exchanges one pair at a time for another signed pair of the catalogue $\mathcal{C}$, accepting any exchange that lowers the misfit, until no single exchange improves it. If the misfit is still too large, the second stage builds a train from scratch, adding one signed pair at a time as in matching pursuit [13] and keeping the W lowest-misfit partial trains at each of the $\widetilde{M}/2$ stages instead of only the best one [14, Sec. 13.4]. The train with the smallest misfit is returned. Algorithm 1 outlines the steps concretely. The obtained $\hat{\mathbf{q}}$ is scaled by 2λ , summed into the staircase $\hat{r}$, and added to y , thus recovering $\hat{g} = y + \hat{r}$.

4.4 Algorithm

Algorithm 1 condenses the procedure of Section 4.3. Default values of the gate and search parameters are listed in Table 4.1. A short numerical example of the catalogue $\mathcal{C}$ and the candidate set $\mathcal{Q}$ is worked through in Appendix A.

The division of labour between the seed obtained by (4.8), and the search to minimise (4.9), follows from the quality of the direct read. When (3.11) holds with margin, the read returns the correct fold vector, and the search does not run. When noise or sampling effects corrupt entries of $\hat{\mathbf{q}}$, for instance a jump displaced by one sample or an exit paired with the wrong re-entry, the leftover in $\mathbf{z}$ still carries the signature of those mistakes. That is the case the fallback addresses. It keeps the estimated fold count $\widetilde{M}$ and relocates a small number of pairs within $\mathcal{Q}$ to minimise the misfit ρ .

Errors in the fold count itself lie beyond the reach of the algorithm, since every candidate in $\mathcal{Q}$ inherits $\widetilde{M}$ from the read. Once λ shrinks toward the per-sample motion of g , the read undercounts folds and no rearrangement of pairs can compensate. The operating boundary of the procedure is

to $\hat{\mathbf{q}} - \mathbf{q} = \pm \mathbf{e}_k$, producing from (4.6) $\mathbf{u} = \pm 2\lambda \mathbf{v}_k + \mathbf{w}$. Correlating $\mathbf{u}$ with signature $\mathbf{v}_k$ yields a peak response:

$$\mathbf{v}_k^H \mathbf{u} = \pm 2\lambda \|\mathbf{v}_k\|_2^2 + \mathbf{v}_k^H \mathbf{w}.$$

Because every entry of $\mathbf{v}_k$ has unit modulus, $\|\mathbf{v}_k\|_2^2 = L$ (the number of out-of-band bins). An unmodelled fold adds in phase to create a coherent peak of magnitude $2\lambda L$, whereas channel noise adds incoherently. The decision threshold $\max_k |\mathbf{v}_k^H \mathbf{u}| < \lambda L$ sits at half that peak, so a fold whose signature has been weakened by noise still triggers the search. We do this deliberately to err on the side of caution because a fold wrongly declared absent corrupts every later sample through the cumulative sum.

Algorithm 1 Fold-sequence recovery by closed-pair search.**Input:**

Folded samples $\{y[k]\}_{k=0}^{K-1}$, $y[k] = \mathcal{M}_\lambda(g[k])$.
 Threshold λ .
 Reserved harmonic count P .
 Cue thresholds η_g, η_j , dilation d , and pair width bound $w_{\max}$.
 Acceptance fraction θ .
 Search width W .

Result:

Jump sequence $\hat{\mathbf{q}}$
 Fold sequence $\hat{r}$
 Unfolded record $\hat{g} = y + \hat{r}$

- 1: Form $(\Delta \mathbf{y})[k] = y[k+1] - y[k]$.
- 2: From $\Delta \mathbf{y}$, get $\hat{\mathbf{q}}$ and count $\tilde{M}$ via (4.8). If $\tilde{M}$ is odd, stop and return y unchanged; the closed-pair model does not apply.
- 3: Build $\mathbf{z}$ as (4.5).
- 4: Build index set $\mathcal{G}$ (4.2) and build the pair catalogue $\mathcal{C}$ (4.3).
- 5: Split $\hat{\mathbf{q}}$ into unit jumps and pair each with the nearest later jump of opposite sign within $w_{\max}$. If the paired read lies in $\mathcal{Q}$ (4.4), keep it as the seed. Otherwise go to step 7.
- 6: Form the spectral residual vector of the seed, $\mathbf{u} = \mathbf{z} + 2\lambda \mathbf{S} \mathbf{V} \hat{\mathbf{q}}$ (see the footnote on page 32), whose norm is the misfit $\rho(\hat{\mathbf{q}})$ (4.7). Accept the seed if $\rho(\hat{\mathbf{q}}) \leq \theta \|\mathbf{z}\|_2$. Otherwise correlate $\mathbf{u}$ with the unit-jump signature $\mathbf{v}_k$ (4.6) at every gap. A fold missed at gap k would leave $2\lambda \mathbf{v}_k$ in $\mathbf{u}$, giving a correlation of $2\lambda L$ because $\|\mathbf{v}_k\|_2^2 = L$, so accept the seed when the largest correlation stays below half that value,

$$\max_k |\mathbf{v}_k^H \mathbf{u}| < \lambda L,$$

with $(\cdot)^H$ the conjugate transpose. If neither test passes, go to step 7.

- 7: Approximate (4.9) over trains of $\tilde{M}/2$ signed pairs from $\mathcal{C}$. Refine the seed by replacing one pair at a time with whichever signed pair of $\mathcal{C}$ lowers ρ , until no replacement improves. If there is no seed, or the refined misfit still exceeds $\theta \|\mathbf{z}\|_2$, grow trains from scratch, keeping the W lowest-misfit partial trains at each stage, and refine the winner the same way. Keep the train with the smallest ρ .
- 8: Scale the accepted $\hat{\mathbf{q}}$ by 2λ and sum cumulatively into the staircase $\hat{r}$ by (3.10), with $\hat{r}[0] = 0$, then return $\hat{g} = y + \hat{r}$ by (3.5).

therefore set by (3.11). The fallback only widens the tolerance to isolated read errors (in terms of positions, widths, signs or pairing) inside that boundary, whether they arise from channel noise, from the rounding of the converter, or from a wrap of the wrong height. Chapter 6 maps the boundary empirically.

Figure 4.5 traces the procedure on one echo of the point-target line, from the difference of the folded samples through to the unfolded record.

4.5 Implementation Notes

The procedure of Algorithm 1 is implemented as a single routine, applied independently to each RF record. The RF data are simulated with the Field II program [4], [5], and the centred-modulo folding and measurement noise are applied to those records in software. The reserved harmonic count P is chosen data-driven as the highest DFT bin at which the spectrum of the record's

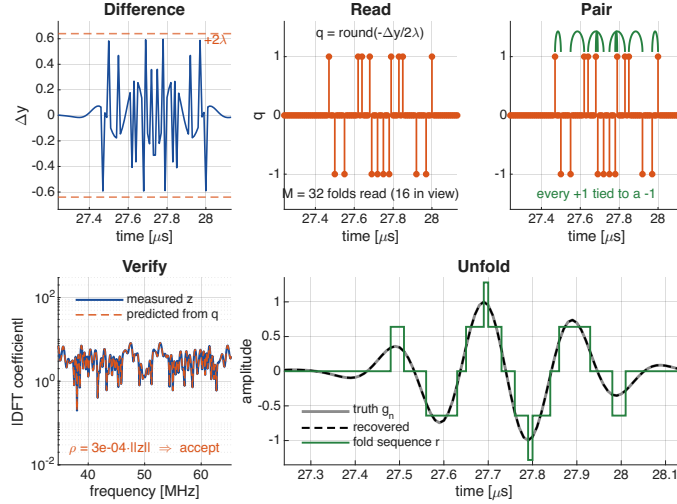


Figure 4.5: The procedure of Algorithm 1 on the brightest echo of the point-target line 17 at $FF = 0.32$ and 80 dB channel SNR. *Difference*, the first difference $\Delta \mathbf{y}$ of the folded samples, with the wrap height $\pm 2\lambda$. *Read*, the direct read of (4.8), which returns ± 1 at every. *Pair*, each $+1$ paired with the nearest later -1 , giving the closed pairs of (4.4). *Verify*, The measured out-of-band spectrum $\mathbf{z}$ is plotted together with the prediction $-2\lambda \mathbf{S} \mathbf{V} \hat{\mathbf{q}}$ from the read. The two curves lie on top of one another, so the misfit (4.7) is 3×10^{-4} of $\|\mathbf{z}\|_2$ and the read is accepted without a search. *Unfold*, the fold sequence rebuilt by cumulative summation and added to the folded samples, which returns the recovered record onto the noisy truth g_n .

brightest line still exceeds a fixed relative floor (10^{-12} of the spectral peak), so that the bins collected into $\mathbf{z}$ carry fold signatures and noise but no pulse leakage (Fig. 4.2). The full data pipeline, noise model, and parameter sweeps are specified in Chapter 5.

Table 4.1 lists the search parameter values used in every experiment reported in Chapter 6.

Table 4.1: Search parameter values used in all reported experiments.

Symbol	Role	Value
η_g	rail cue, flag $ y > \eta_g \lambda$	0.85
η_j	jump cue, flag $ \Delta y > \eta_j \lambda$	1.0
d	dilation of flagged indices, $\pm d$ samples	3
$w_{\max}$	widest admissible pair	15
θ	seed acceptance, $\rho \leq \theta \ \mathbf{z}\ _2$	0.05
W	partial candidates kept per stage of the search	32

The bound $w_{\max}$ is set by how long an echo can spend outside $\pm \lambda$. That interval cannot exceed half a period of the centre frequency (Fig. 3.1), or 10 samples at $f_s = 100$ MHz and $f_c = 5$ MHz. The value used here is 15, which leaves margin above that physical bound.

When the catalogue $\mathcal{C}$ grows beyond roughly 2×10^4 pairs the search stage of step 7 is skipped and the seed answer is kept, since the signature matrix would exceed practical memory requirements.

4.6 Summary

This chapter has explained the recovery problem, shown why the spectral baseline fails on the paired folds of ultrasound RF, and set out a procedure that reads the fold sequence directly, verifies it against the spectral coefficients outside the RF bandlimit, and searches for a better fold sequence when the read is dubious. Chapter 5 specifies how the algorithm was tested, the simulated data and noise model, the baselines for comparison, and the evaluation metrics. Chapter 6 then reports the outcomes and maps the operating boundary anticipated in Section 4.3.

Experimental Design

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The following describes the experiments carried out to test the recovery procedure described in Chapter 4. Algorithm 1 is tested on simulated ultrasound RF. The experiments evaluate the recovery achieved by the algorithm under Gaussian noise, fold jitter, and finite-bit quantisation.

5.1 Signal Pipeline

Figure 5.1 shows the experimental setup. Field II simulates the ultrasound RF signal g , to which Gaussian noise is added to produce g_n . The centred modulo operator at threshold λ then generates y , and algorithm 1 reconstructs the estimated signal $\hat{g}$.

This chain is applied in two scenarios: where g is a beamformed A-line, and where g is a raw receive-channel record. In the latter case, folding and recovery are applied independently to each of the 64 active channels across all 129 scan lines before delay-and-sum beamforming creates the final B-mode image.

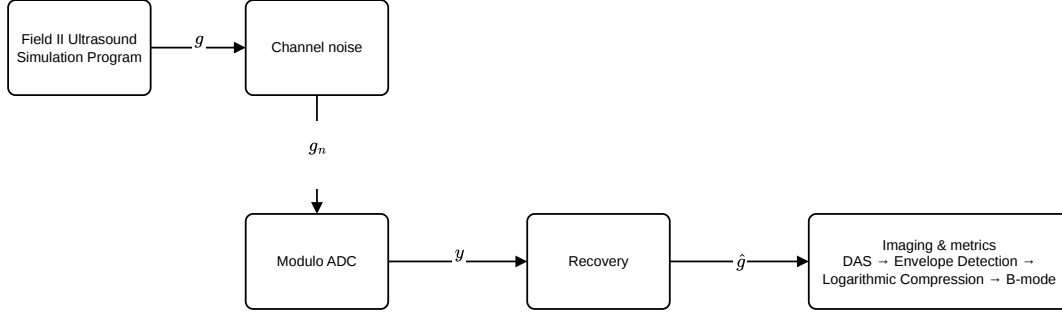


Figure 5.1: Processing chain used in all tests. Field II provides the ground-truth RF signal g , to which Gaussian noise is added to form g_n . A centred modulo map at threshold λ produces y , and algorithm 1 reconstructs $\hat{g}$. Here, g represents either a beamformed A-line or a single receive-channel record. The DAS, envelope, and log-compression stages apply only to per-channel recovery.

5.2 Data and Simulation Setup

5.2.1 Phantom and Ground Truth

The phantom is a 3D box measuring 50 mm in width (x), 10 mm in height (y), and 60 mm in depth (z), positioned 30 mm from the transducer face. It contains 10,000 scatterers distributed uniformly throughout the volume. To generate background speckle, the scattering amplitudes are modelled as independent and identically distributed draws from a standard normal distribution $\mathcal{N}(0,1)$. Hyperechoic regions, anechoic cysts, and point targets are introduced by modifying those amplitudes.

- Hyperechoic regions have amplitudes scaled by a factor of 10.
- Anechoic cysts have amplitudes set to zero.
- Point targets are assigned a fixed deterministic amplitude of 20.

Table 5.1 lists the embedded structures and their corresponding generation parameters. The background scatterers are generated using a fixed random seed to ensure exact reproducibility across realisations.

Figure 5.2 shows the Field II cyst phantom used throughout this study [17].

Table 5.1: Structures embedded in the Field II cyst phantom (Fig. 5.2). Diameters are listed in order of increasing depth (z). Point targets have fixed deterministic amplitudes; anechoic cysts contain no scatterers; and hyperechoic regions scale the background speckle amplitudes by a factor of 10.

Structure	N	x (mm)	z (mm)	Diameter (mm)	Scatterer strength
Point targets	6	-15	40, 50, 60, $\dots$, 90	—	20 (fixed)
Anechoic cysts	5	+10	40, 50, 60, 70, 80	6, 5, 4, 3, 2	none
Hyperechoic regions	5	-5	40, 50, 60, 70, 80	1, 2, 3, 4, 5	speckle $\times 10$

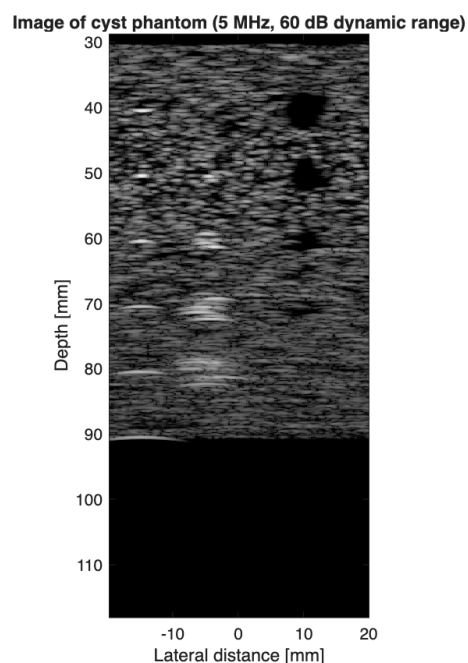


Figure 5.2: B-mode image of the Field II cyst phantom at 5 MHz (60 dB dynamic range). The column of point targets is located at lateral position $x = -15$ mm, hyperechoic regions at $x = -5$ mm, and anechoic cysts at $x = +10$ mm. The background shows uniform speckle texture.

5.2.2 Sampling, folding threshold, and test records

Field II records are simulated at a centre frequency of $f_c = 5$ MHz and a sampling frequency of $f_s = 100$ MHz, with a speed of sound of $c = 1540$ m s⁻¹. The simulation script is adapted from J. A. Jensen's course materials on ultrasound imaging¹. The probe is modelled as a 192-element linear array. Each element has height 5 mm, width $w = c/f_c = 0.308$ mm, and kerf $k = c/(20f_c) = 15.4$ μ m, yielding a pitch of $p = w + k = 0.323$ mm. A 64-element active aperture with a transmit focus at 50 mm is translated across the array to form 129 scan lines with an inter-line spacing of 0.308 mm. Transmit excitation is a two-cycle sinusoid at f_c , and both transmit and receive apertures use Hanning-weighted impulse responses. Tissue attenuation is omitted.

A-line recovery is evaluated on three representative beamformed lines selected by crest factor, $\frac{\max |g|}{\text{RMS}(g)}$, to span the key signal regimes of ultrasound RF (Fig. 5.3). Line 17 represents a **point-target line** ($x = -15$ mm, crest factor 17.4) characterised by strong discrete echoes. Lines 37 and 75 represent **speckle-dominated lines** (crest factors 6.0 and 6.2) taken from high-energy regions without dominant reflectors.

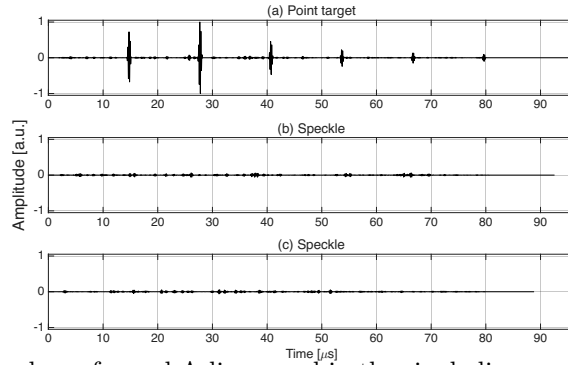


Figure 5.3: Representative beamformed A-lines used in the single-line recovery evaluation, plotted on a common amplitude scale: (a) Point-target Line 17 (crest factor 17.4); (b)–(c) Speckle Lines 37 and 75 (crest factors 6.0 and 6.2). The common scale is for visual comparison only; recovery sets λ dynamically per line.

For a single A-line, the folding threshold is set relative to its peak amplitude, $\lambda = \text{FF} \cdot \max_k |g[k]|$. The fold factor (FF) denotes the fraction of the peak covered by the dynamic range of the ADC; for example, $\text{FF} = 0.35$ means the threshold spans 35% of $\max |g|$. When evaluating per-channel recovery across a scan line, a unified threshold $\lambda = \text{FF} \cdot \max |g|$ is calculated across all 64 active channels. Each channel is folded and reconstructed independently using this shared threshold, simulating array elements operating with identical ADC input ranges.

¹Exercise material for *Advanced Ultrasound Imaging* (J. A. Jensen, Technical University of Denmark, 2025), https://home.healthtech.dtu.dk/jaj/ai_2025/; https://field-ii.dk/examples/ftp_files/cyst_phantom/.

5.3 Sweep and Monte Carlo Protocol

The performance of Algorithm 1 is evaluated across two test regimes using the core processing chain shown in Fig. 5.1. The first regime assesses single-line recovery using the three representative beamformed A-lines from Fig. 5.3. The second regime performs channel-level recovery across all receive channels prior to B-mode image formation. All experiments use a fixed scatterer phantom realisation. The A-line evaluations additionally incorporate fold jitter and finite-bit quantisation.

5.3.1 A-line tests

For each A-line, the folding threshold is fixed from the clean peak, $\lambda = \text{FF} \cdot \max_k |g[k]|$, ensuring that FF does not drift with noise. Each Monte Carlo draw follows a fixed chain: pre-fold additive noise yields g_n (Eq. 3.2), the centred modulo map folds g_n into $y \in [-\lambda, \lambda]$ (Eq. 3.4), fold jitter optionally perturbs the wrap heights (Section 3.2.4), and a finite-bit quantiser optionally rounds the resulting signal (Section 3.1.2). Algorithm 1 receives y and λ to reconstruct $\hat{g}$.

Noise. Pre-fold channel noise is simulated as additive white Gaussian noise following Eq. (3.2). The noise variance σ is scaled relative to the root-mean-square (RMS) level of the full clean record g (Eq. 3.3), making the nominal channel SNR exact by construction. Under this formulation, a perfect recovery returning g_n yields a baseline NRMSE of $10^{-\text{SNR}/20}$ relative to g .

Fold jitter. Fold-amplitude jitter is modelled as described in Section 3.2.4. Each wrap event $q \in \mathbb{Z}$ is perturbed to $2\lambda(q + \varepsilon)$, where $\varepsilon \sim \mathcal{U}(-\varepsilon_{\max}, +\varepsilon_{\max})$ is drawn independently per fold. The jitter bound $\varepsilon_{\max}$ is quoted throughout as a percentage of 2λ .

Quantisation. When enabled, the folded signal (including any jitter perturbation) is rounded to B bits on $[-\lambda, \lambda]$ with step size $\Delta = 2\lambda/2^B$, following the quantiser model in Section 3.1.2.

Scoring and Monte Carlo execution. The reference fold sequence $\mathbf{q}$ is extracted from the noisy, unjittered, unquantised record g_n . Non-idealities (jitter and quantisation) are not permitted to redefine the ground-truth target. NRMSE is evaluated against g_n ; thus, under jitter, an exact fold train recovery means the true jump integers were identified, while reconstructed amplitudes retain the physical wrap error.

Table 5.2 details the parameter combinations tested per A-line. Each combination evaluates $N = 50$ independent noise realisations. The fold count M is estimated dynamically from the data across all runs. To isolate parameter effects, identical noise seeds are reused across fold factors, bit depths, and jitter settings.

Table 5.2: Parameter sweep grids for single-line recovery tests across Line 17 (point target), Line 37 (speckle), and Line 75 (speckle). Baseline settings when unvaried are SNR = 80 dB, quantisation = off (–), and jitter = off (–). All runs evaluate $N = 50$ noise draws per grid point.

Sweep	Target Lines	FF	SNR (dB)	B (bits)	Jitter (% of 2λ)
Threshold	All	0.32 to 0.90 (step 0.10)	80	–	–
Boundary	All	0.28, 0.29, 0.30	80	–	–
Noise	Line 17	0.32, 0.35, 0.38, 0.40	10, 20, 30, 40, 60, 80	–	–
	Lines 37, 75	0.30, 0.35, 0.40	10, 20, 30, 40, 60, 80*	–	–
Quantisation	Line 17	0.32, 0.40	80	2 to 8, 10, 12	–
	Lines 37, 75	0.30, 0.40	80	2 to 8, 10, 12	–
Jitter	Line 17	0.32, 0.60	80	–	0, 1, 2, 5, 10, 20
	Lines 37, 75	0.30, 0.60	80	–	0, 1, 2, 5, 10, 20
Jitter Ref.	Lines 17, 75	0.40	80	–	2, 5
Combined	Lines 17, 75	0.40	20, 40	8, 10	2, 5

* Line 37 noise sweep evaluates SNR $\in \{20, 40, 60, 80\}$ dB.

5.3.2 Per-channel B-mode tests

To evaluate performance at the system level, recovery is performed on each raw receive channel prior to delay-and-sum beamforming. Six B-mode images are constructed across a parameter grid of $\text{FF} \in \{0.35, 0.50\}$ and channel SNR $\in \{\text{clean}, 20 \text{ dB}, 40 \text{ dB}\}$. Each image comprises 129 scan lines across 64 active channels, totaling 8,256 independent channel reconstructions per image. A fixed random noise seed is used per image state.

For each scenario, processing is conducted on the same folded data using a shared per-line folding threshold, $\lambda = \text{FF} \cdot \max |g|$, computed across that line’s 64 channels. Four distinct processing paths are compared:

1. **Proposed modulo path:** Folds and reconstructs each channel independently prior to beamforming.
2. **Unconstrained reference:** Beamforms the unquantised, clean RF signal (no-ADC baseline).
3. **Saturating reference:** Applies hard clipping at $\pm\lambda$ per channel prior to beamforming.

4. **Post-beamform contrast:** Folds and recovers the already beamformed A-line (included purely for comparison, not as a primary hardware model).

Image-level evaluation metrics and baseline criteria are defined in Section 5.4.

5.4 Baselines, Metrics, and Success Criteria

To evaluate algorithm performance across single-line and image-level experiments, reconstructions are assessed using both structural correctness of the fold sequence and quantitative amplitude errors.

5

5.4.1 Reference Baselines

Reconstructions are benchmarked against three reference representations:

1. **A-line amplitude baseline:** The noisy pre-fold signal g_n . Evaluating error against g_n ensures that additive noise is treated as an expected component of the target signal rather than a reconstruction error.
2. **Ground-truth fold train:** The reference fold sequence $\mathbf{q}$ extracted from g_n prior to quantisation or fold jitter (Section 5.3.1).
3. **B-mode image baselines:** Beamformed B-mode images constructed from (i) unconstrained, full-dynamic-range RF signals (ground truth), and (ii) hard-clipped RF signals bounded at $\pm\lambda$.

5.4.2 Metrics and Success Criteria

A recovery run is classified as successful if the estimated fold sequence $\hat{\mathbf{q}}$ matches the ground-truth sequence $\mathbf{q}$ in both fold locations and integer wrap amplitudes across the entire record length.

Quantitative reconstruction quality is evaluated using normalised root-mean-square error (NRMSE) relative to g_n :

1. **Single A-line metrics:** For ideal and noise-only runs, a successful recovery must achieve $\text{NRMSE} \leq 1\%$. When fold jitter is active, correct identification of $\mathbf{q}$ is still required for

success; however, residual NRMSE may exceed 1% due to physical wrap-amplitude perturbations inherent to the jittered signal model. Performance across Monte Carlo sweeps is reported via the exact-recovery success rate (fraction of $N = 50$ trials) and median NRMSE.

2. **B-mode image metrics:** For channel-level evaluation, a line is considered successful only when every active receive channel (64 channels) achieves exact fold-train recovery. Reconstructed B-mode images are evaluated qualitatively and quantitatively against the ground-truth and hard-clipped reference images.

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This chapter presents the experimental evaluation of the modulo signal recovery pipeline introduced in Chapter 4, following the methodology detailed in Chapter 5. The analysis addresses four core research questions:

1. **Beamformed A-line performance:** Does the algorithm achieve exact fold-sequence recovery on single A-lines, and what are the precise folding-fraction (FF) limits of this operating region?
2. **Non-ideal hardware robustness:** Does the stable operating region persist under the compound hardware non-idealities modelled in Section 5.3.1, specifically additive thermal noise, ADC bit-depth quantisation, and fold-edge jitter?

3. **Channel-level B-mode fidelity:** When modulo acquisition is performed per receive channel prior to delay-and-sum beamforming, how closely does the reconstructed B-mode image match the unconstrained ground-truth reference?
4. **Comparison to saturation:** How does the proposed channel-level modulo recovery compare against standard hard clipping at equivalent dynamic range boundaries ($\pm\lambda$)?

6.1 Fold-Sequence Recovery on A-lines

Baseline fold-recovery performance was evaluated on the three representative beamformed A-lines under nominal, high SNR hardware conditions (SNR = 80 dB, unquantised, zero jitter). For each folding fraction (FF), fold counts were estimated directly from the data via Algorithm 1 across $N = 50$ independent pre-fold noise realisations.

Across all recovering configurations, the algorithm achieved 100% exact fold-train recovery (50/50 trials) and perfect fold-count estimation ($\widetilde{M} = M$). As summarised in Table 6.1, exact recovery held down to FF = 0.32 for the point-target record (Line 17, $M = 32$) and down to FF = 0.30 for the speckle records (Line 37, $M = 256$; Line 75, $M = 234$). Below these critical thresholds, recovery fails completely due to first-order difference violation.

Quantitatively, the median NRMSE relative to the uncorrupted input g is 0.01%, matching the noise floor of the 80 dB input channel. When evaluated against the true folded input g_n , the median NRMSE sits at machine precision ($\approx 10^{-14}\%$). Figure 6.1 illustrates this exact convergence on representative excerpts at the densest passing settings (FF = 0.32 for Line 17 and FF = 0.30 for Line 37), where the reconstructed signal $\hat{g}$ overlays g_n perfectly.

The breakdown on Line 17 at FF = 0.30 is directly governed by a violation of Itoh's condition (Eq. (3.11)). When the sample-to-sample first-order difference exceeds λ , the estimated fold vector read from the modulo signal (Eq. (4.8)) incurs a step error, causing the reconstructed staircase sequence $\hat{r}$ to shift by an integer multiple of $\pm 2\lambda$ from that sample onward. Formally, this constraint establishes a lower bound on the folding fraction, FF_{bound} :

$$\text{FF}_{\text{bound}} = \frac{\max_k |(\Delta g)[k]|}{\max_k |g[k]|}. \quad (6.1)$$

Because the speckle records possess lower maximum derivative magnitudes compared to the point-target echo, Lines 37 and 75 maintain valid modulo unwrapping conditions down to FF = 0.30

Table 6.1: Fold-sequence recovery across three beamformed A-lines at 80 dB channel SNR ($N = 50$ noise realisations, unquantised, zero jitter). Across all successful runs, exact fold-train recovery was achieved in 50/50 trials ($\tilde{M} = M$), with median NRMSE against clean g at 0.01%. Tabulated NRMSE values represent median errors computed against noisy input g_n .

FF	Line 17 (Point Target)		Line 37 (Speckle)		Line 75 (Speckle)	
	M	NRMSE [$\times 10^{-14}\%$]	M	NRMSE [$\times 10^{-14}\%$]	M	NRMSE [$\times 10^{-14}\%$]
0.90	4	5.69	14	2.36	8	2.48
0.80	4	5.65	26	2.16	14	2.24
0.70	10	5.63	44	2.07	32	2.12
0.60	12	5.61	70	1.98	50	2.02
0.50	14	4.45	104	1.64	92	1.68
0.40	20	2.89	156	1.25	150	1.29
0.32	32	2.89	226	1.24	210	1.24
0.30		Fails	256	1.22	234	1.21
0.29		Fails	276	1.20		Fails
0.28		Fails		Fails		Fails

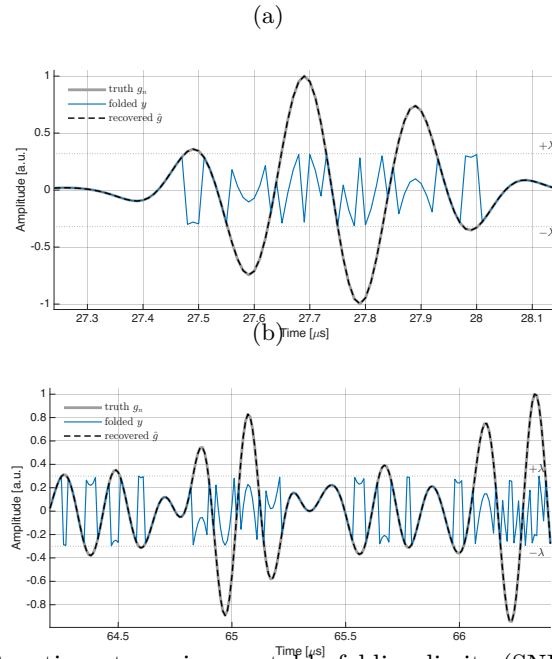


Figure 6.1: Signal reconstruction at maximum stable folding limits (SNR = 80 dB): (a) Line 17 excerpt centred on the dominant point-target echo (FF = 0.32); (b) Line 37 excerpt representing the maximum fold density record (FF = 0.30, $M = 256$). The reconstructed signal $\hat{g}$ overlays the noisy truth g_n exactly in both cases.

and 0.29, respectively.

Crucially, the reciprocal of the folding fraction quantifies the dynamic range expansion achieved relative to a conventional ADC bounded at $\pm\lambda$. Both converters digitise within $\pm\lambda$ at B bits, so both have the same quantisation step $\Delta = 2\lambda/2^B$. The conventional converter clips at λ , whereas recovery reconstructs peaks up to λ/FF , so

$$\frac{\text{DR}_{\text{mod}}}{\text{DR}_{\text{conv}}} = \frac{(\lambda/\text{FF})/\Delta}{\lambda/\Delta} = \frac{1}{\text{FF}} \quad \text{or} \quad 20 \log_{10} \frac{1}{\text{FF}} \text{ dB}. \quad (6.2)$$

At $\text{FF} = 0.32$ for Line 17 and $\text{FF} = 0.30$ for speckle, the reconstructed peak amplitudes reach 3.1λ and 3.3λ , a dynamic range extension of approximately 10 dB.

Having established the fundamental mathematical stability boundary under nominal, high-SNR conditions ($\text{SNR} = 80$ dB, unquantised, zero jitter), Section 6.2 evaluates the resilience of this operating window under realistic hardware impairments.

6.2 Robustness to Noise, Quantisation, and Jitter

To evaluate algorithmic robustness under non-ideal hardware conditions, the three imperfections modelled in Section 5.3.1—additive thermal noise, finite-bit quantisation, and fold-edge jitter—were evaluated individually and in combination across the three representative A-lines.

Fundamentally, algorithm stability depends on maintaining a positive difference margin m . When operating above the minimum folding bound ($\text{FF} > \text{FF}_{\text{bound}}$), the remaining capacity before violating Itoh’s condition defines the read margin:

$$m = \lambda - \max_k |(\Delta g)[k]| = \lambda \left(1 - \frac{\text{FF}_{\text{bound}}}{\text{FF}} \right). \quad (6.3)$$

This margin m represents the maximum allowable amplitude perturbation—contributed jointly by additive noise, quantisation rounding, and fold-height variations—that the algorithm can tolerate before misidentifying the fold vector $\mathbf{q}$.

6.2.1 Channel Noise

Table 6.2 details exact fold-sequence recovery success rates across varying folding fractions (FF) and channel SNRs. Across all successful trials, the reconstruction error relative to the clean signal g is governed strictly by the additive noise floor, yielding a median NRMSE equal to $10^{-\text{SNR}/20}$ (ranging from 31.6% at 10 dB to 0.01% at 80 dB). Relative to the noisy input g_n , the error remains at machine precision.

Table 6.2: Exact fold-train recovery success rates out of $N = 50$ trials across channel SNRs and folding fractions (FF). Normalised read margin m/λ indicates the residual error tolerance prior to violating Itoh’s condition.

Line (FF)	m/λ	Channel SNR (dB)					
		10	20	30	40	60	80
17 (0.40)	21.9%	50	50	50	50	50	50
17 (0.38)	17.8%	50	50	50	50	50	50
17 (0.35)	10.8%	43	50	50	50	50	50
17 (0.32)	2.4%	8	33	50	50	50	50
37 (0.40)	28.9%	7	50	50	50	50	50
37 (0.35)	18.7%	0	49	50	50	50	50
37 (0.30)	5.2%	0	4	48	50	50	50
75 (0.40)	26.2%	18	50	50	50	50	50
75 (0.35)	15.7%	1	48	50	50	50	50
75 (0.30)	1.7%	0	8	31	48	50	50

The failure mechanisms observed at reduced SNRs originate in the direct fold-vector read phase of Algorithm 1:

1. **Rounding bit-flips:** High-amplitude noise perturbations on adjacent modulo samples y can cause the nearest-integer operator in Eq. (4.8) to misidentify a fold location or amplitude step.
2. **Parity violations:** Because signal excursions outside $\pm\lambda$ must enter and exit in pairs, an isolated read error yields an odd fold count $\widetilde{M}$, which the algorithm rejects.
3. **Combinatorial search truncation:** Under high fold densities (e.g., Line 37 at $\text{FF} = 0.30$, $M = 256$) and low SNRs, the number of admissible candidate fold pairs exceeds the search limit of the algorithm (2×10^4). In these heavily corrupted cases, global combinatorial search is bypassed to constrain computation times, leaving uncorrected initial read errors.

6.2.2 Finite-Bit Quantisation

A B -bit analogue-to-digital converter bounded on $[-\lambda, \lambda]$ rounds each modulo sample y , introducing a deterministic quantisation error bounded by $\|\eta\|_\infty \leq \lambda 2^{-B}$. Bhandari et al. [2] show that an N th-order difference inflates such error by at most 2^N . For the first-order differences ($N = 1$) used in the direct read phase, quantisation expands adjacent sample errors by up to $\lambda/2^{B-1}$. To ensure exact fold-vector recovery, this maximum quantisation error must remain strictly within the available read margin m (Eq. (6.3)):

$$\frac{m}{\lambda} > \frac{1}{2^{B-1}}. \quad (6.4)$$

Table 6.3 summarises exact recovery performance across bit resolutions at SNR = 80 dB. As predicted by Eq. (6.4), the minimum bit resolution required for exact fold-train reconstruction depends directly on m/λ : wide margins (FF = 0.40) recover reliably at $B = 3$ bits, whereas operating near slope boundaries (FF = 0.30) requires up to $B = 7$ bits.

Table 6.3: Exact fold-sequence recovery ($N = 50$ trials, SNR = 80 dB, zero jitter) across converter bit depths B . Normalised read margin m/λ governs the minimum resolution required for exact reconstruction.

Line (FF)	m/λ	B (bits)							
		2	3	4	5	6	7	8	10
17 (0.40)	21.9%	0	50	50	50	50	50	50	50
37 (0.40)	28.9%	0	50	50	50	50	50	50	50
75 (0.40)	26.2%	0	50	50	50	50	50	50	50
37 (0.30)	5.2%	0	0	0	50	50	50	50	50
17 (0.32)	2.4%	0	0	0	0	0	50	50	50
75 (0.30)	1.7%	0	0	0	0	50	50	50	50

Because the reconstructed signal is formed as $\hat{g} = y_q + \hat{r}$ (where y_q is the quantised modulo signal), exact identification of the fold sequence $\hat{r}$ leaves residual quantisation noise in y_q [2]. Table 6.4 evaluates reconstruction errors across typical clinical ADC bit depths. Records with high crest factors (such as the point-target Line 17) exhibit higher relative NRMSE for a given bit depth compared to speckle-dominated signals.

6.2.3 Fold-Edge Jitter

To evaluate hardware fold-height non-idealities, fold-edge jitter was introduced following Section 5.3.1. Because the recovery architecture assumes ideal 2λ modulo resets, amplitude variation

Table 6.4: Median NRMSE of $\hat{g}$ against g_n across representative clinical ADC bit depths ($N = 50$ trials, exact fold recovery achieved across all entries).

Line (FF)	Crest Factor	$B = 8$	$B = 10$	$B = 12$
17 (0.40)	17.4	1.40%	0.359%	0.0898%
17 (0.32)	17.4	1.12%	0.285%	0.0719%
37 (0.40)	6.0	0.500%	0.124%	0.0315%
37 (0.30)	6.0	0.374%	0.0943%	0.0240%
75 (0.40)	6.2	0.526%	0.133%	0.0337%
75 (0.30)	6.2	0.399%	0.101%	0.0252%

in the resets introduces a structural model mismatch. Table 6.5 evaluates fold-train recovery across jitter severities at SNR = 80 dB (unquantised), expressed relative to the full reset height 2λ .

Fundamentally, a fold step of non-ideal height introduces an amplitude error bounded by $2\lambda\epsilon_{\max}$. The direct fold read successfully recovers the integer sequence as long as this perturbation remains within the normalised read margin:

$$\epsilon_{\max} < \frac{m}{2\lambda}. \quad (6.5)$$

Table 6.5: Exact fold-sequence recovery success rates ($N = 50$ trials, SNR = 80 dB, unquantised) across jitter severities expressed as a percentage of 2λ . Read margins $m/2\lambda$ represent the normalised threshold before violating Itoh's condition.

Line (FF)	$m/2\lambda$	Jitter Severity (% of 2λ)					
		0	1	2	5	10	20
17 (0.60)	24.0%	50	50	50	50	50	50
37 (0.60)	26.3%	50	50	50	50	50	50
75 (0.60)	25.4%	50	50	50	50	50	50
37 (0.30)	2.6%	50	50	50	36	8	0
17 (0.32)	1.2%	50	50	41	29	28	16
75 (0.30)	0.8%	50	50	39	29	16	1

Crucially, as stated in previous instances, under fold jitter, exact fold-train identification does not imply zero reconstruction error. Because the recovery algorithm assumes nominal $2\lambda\mathbf{q}$ steps, the jitter perturbations $2\lambda\epsilon$ introduced during folding remain uncompensated in the reconstructed record $\hat{g}$. This residual error scales linearly with jitter severity; for instance, on Line 17 (FF = 0.60), the median NRMSE against g_n increases linearly at approximately 8.6% per 1% jitter severity (8.6%, 17.2%, and 43% NRMSE at 1%, 2%, and 5% severity, respectively). A refit to arbitrary wrap heights is left as future work.

6.2.4 Combined Hardware Stressors

To evaluate algorithmic resilience under operational conditions, additive noise, finite-bit quantisation, and fold jitter were applied simultaneously at $\text{FF} = 0.40$ across all three A-lines. Parameter combinations spanned $\text{SNR} \in \{20, 40\}$ dB, quantisation resolution $B \in \{8, 10\}$ bits, and jitter severity $\varepsilon \in \{2\%, 5\%\}$ of 2λ . Across all 24 evaluated parameter combinations ($N = 50$ trials per cell), the algorithm achieved 100% exact fold-train recovery.

Table 6.6: Median NRMSE against g_n at $\text{FF} = 0.40$ under combined hardware stressors ($N = 50$ trials, exact fold recovery achieved in 100% of cases). Reference row presents performance under jitter alone at the same folding fraction ($\text{SNR} = 80$ dB, unquantised).

SNR (dB)	B	Line 17 (Point Target)		Line 37 (Speckle)		Line 75 (Speckle)	
		Jitter 2%	Jitter 5%	Jitter 2%	Jitter 5%	Jitter 2%	Jitter 5%
40	10	13.1%	32.6%	16.4%	40.9%	14.2%	35.6%
40	8	13.2%	32.7%	16.4%	40.9%	14.3%	35.6%
20	10	14.1%	35.3%	14.8%	37.0%	14.4%	36.1%
20	8	14.2%	35.3%	14.8%	37.0%	14.5%	36.1%
Jitter alone (80 dB, unquantised)		15.2%	38.0%	15.3%	38.3%	13.7%	34.3%

As summarised in Table 6.6, the total residual error is dominated almost entirely by fold-jitter amplitude variations ($2\lambda\varepsilon$) rather than additive noise or quantisation rounding. Whereas standalone noise leaves near-zero error against g_n and standalone 8-bit quantisation contributes $\approx 0.5\%$ – 1.4% NRMSE, jitter at 2% and 5% severities imposes baseline error floors of $\approx 14\%$ and 36% , respectively. Provided the combined perturbation magnitude does not exceed the available read margin m , integer fold identification $\hat{q}_k$ remains fully possible.

Figure 6.2 illustrates this behaviour on Speckle Line 37 under combined stressors ($B = 8$ bits, $\text{SNR} = 20$ dB, 2% jitter) alongside a conventional linear 8-bit ADC. Without modulo folding, the standard ADC clips all peak excursions exceeding $\pm\lambda$, resulting in severe signal distortion (29% NRMSE) and losing 4.6% of recorded samples. In contrast, the modulo architecture recovers the complete continuous-wave structure up to $1/\text{FF} = 2.5\times$ the converter full-scale range—a dynamic-range extension of 8 dB—with residual error (12.3% NRMSE) arising strictly from reset-height jitter.

6.3 Per-Channel Folding and B-mode Images

To evaluate performance within a realistic system architecture (where an ADC digitises individual receive elements prior to beamforming) the modulo recovery procedure was extended to all 64 active

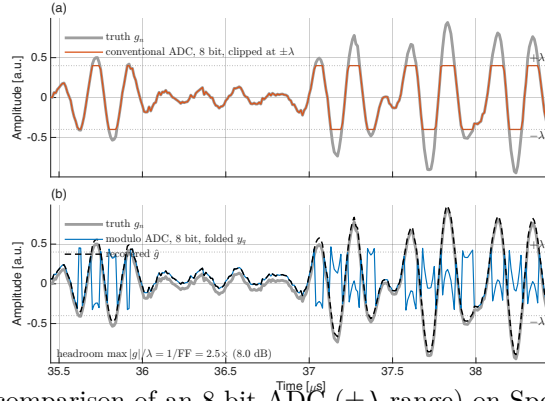


Figure 6.2: Performance comparison of an 8-bit ADC ($\pm\lambda$ range) on Speckle Line 37 ($FF = 0.40$, $SNR = 20$ dB, 2% jitter over a $3\mu s$ excerpt containing 32 folds): (a) Unmodulated ADC severe clipping above $\pm\lambda$; (b) Modulo ADC recording folded samples y_q and reconstructed signal $\hat{g}$, achieving an 8 dB dynamic-range expansion.

receive channels across the 129 transmit scanlines. A threshold $\lambda = FF \cdot \max |g|$, was derived from the peak RF amplitude observed across all 64 channels on a given line. Each channel was folded and reconstructed independently before undergoing delay-and-sum (DAS) beamforming to form the final A-lines and B-mode images.

Figure 6.3 illustrates a representative reconstruction for channel 32 of line 17 at $FF = 0.35$ under 20 dB channel SNR. The folded signal remains strictly within the shared dynamic range $\pm\lambda$, while the reconstructed signal faithfully matches the noisy channel ground truth. Because the processing occurs pre-beamforming, echoes appear at their un-delayed element arrival times.

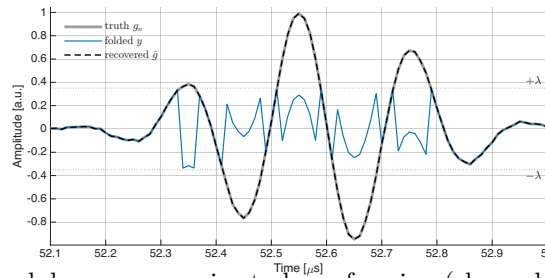


Figure 6.3: Per-channel modulo recovery prior to beamforming (channel 32 of 64 active elements, line 17, $FF = 0.35$, $SNR = 20$ dB, shown over a $0.9\mu s$ window containing 10 of the 26 channel folds). Grey trace: noisy channel record g_n ; blue trace: folded record y constrained within $\pm\lambda$ (dotted); black dashed trace: reconstructed signal $\hat{g}$. The algorithm operates solely on y and λ , achieving an exact integer fold recovery with residual error bounded by machine precision ($2.5 \times 10^{-14}\%$ NRMSE).

Following independent per-channel recovery, the signals are aligned via focusing delays and summed to synthesise the beamformed A-line.

6.3.1 Effect of Delay-and-Sum Beamforming

A scanline formed by Delay-and-Sum (DAS) is a weighted sum of all 64 active receive channel signals, aligned in time so that coherent reflections add constructively across the aperture. Apodisation weighting (a Hamming window in this implementation) tapers the aperture edges to suppress spatial sidelobes.

In the event of an isolated integer recovery failure on a single channel, the resulting trajectory error (a step of height 2λ) enters as a single term within this 64-element sum. Because the remaining 63 channels add coherently, the relative amplitude distortion on the beamformed line is attenuated by a factor proportional to the channel's relative apodisation weight. For channels near the aperture edges, where Hamming weights are smallest, the artefact is suppressed by approximately $1/64 \approx -36$ dB.

6.3.2 B-mode Image Reconstruction Compared with Hard Clipping

To evaluate full image reconstruction, B-mode images were formed by independently folding and recovering all $129 \times 64 = 8256$ channel records at a folding fraction of $FF = 0.35$ across channel noise levels $SNR \in \{\text{clean}, 40, 20\}$ dB. The reconstructed B-mode images were benchmarked against an ideal reference image (no ADC), as well as an image subject to hard clipping at the same dynamic range threshold $\pm\lambda$.

Figure 6.4 presents spatial pixel difference maps at $FF = 0.35$ and 40 dB SNR. While standard hard clipping severely distorts the acoustic structure and background speckle, per-channel modulo recovery eliminates dynamic range clipping artefacts, reconstructing an image identical to the unconstrained reference.

Table 6.7 quantifies single-pixel deviations across operating conditions. At $FF = 0.35$, the folding threshold accommodates signal excursions reaching approximately 2.9λ . Across all clean and 40 dB SNR trials, per-channel recovery reproduces the ideal B-mode image to within machine precision ($|\overline{\Delta I}| \approx 10^{-13}$ dB). At the most punishing operational threshold ($FF = 0.35$, 20 dB SNR), 6 isolated channel failures out of 8,256 (0.07%) introduce a worst-case single-pixel local error of 14.7 dB, though mean image-wide error remains negligible (1.6×10^{-3} dB).

Thus, applying a uniform threshold λ across receive channels, per-channel modulo folding and recovery successfully preserves B-mode image fidelity while overcoming clipping from ADC.

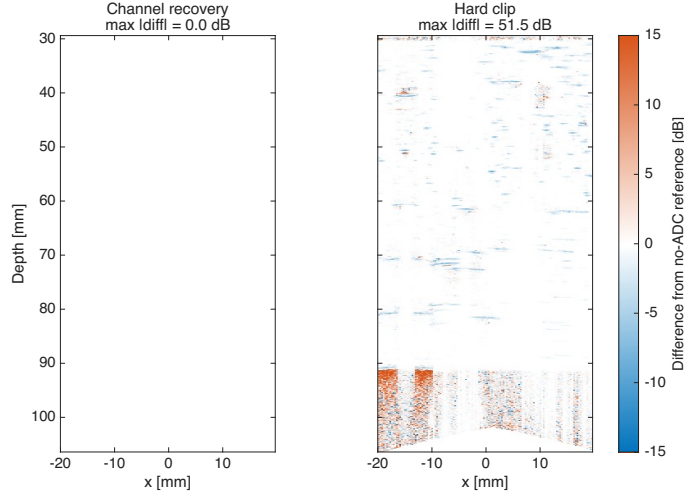


Figure 6.4: Log-compressed pixel brightness difference ($\Delta I = I - I_{\text{ref}}$ in dB) relative to the ideal unconstrained reference image at FF = 0.35 and SNR = 40 dB. Blue indicates pixels darker than the reference; orange indicates pixels brighter. Left: Per-channel modulo folding and recovery. Right: Hard clipping at the identical dynamic range threshold $\pm\lambda$.

Table 6.7: Pixel-by-pixel log-compressed envelope error ($\Delta I = I - I_{\text{ref}}$ in dB) relative to the unconstrained no-ADC reference image. Each row evaluates a full B-mode image (129 scanlines by 10^4 axial samples) synthesised from 8,256 independent channel recoveries. Metrics quote the worst-case pixel error ($\max |\Delta I|$) out of $\approx 1.3 \times 10^6$ pixels and image-wide average error ($|\overline{\Delta I}|$). Hard clipping applies the identical input range $\pm\lambda$ without recovery.

FF	SNR	Channel Recovery (dB)		Hard Clipping (dB)	
		$\max \Delta I $	$ \overline{\Delta I} $	$\max \Delta I $	$ \overline{\Delta I} $
0.35	clean	9.8×10^{-10}	4.0×10^{-13}	50.6	2.80
0.35	40 dB	2.2×10^{-10}	3.7×10^{-13}	51.5	0.620
0.35	20 dB	14.7	1.6×10^{-3}	38.6	0.221

6.4 Discussion

Applying the Unlimited Sensing Framework (USF) developed by Bhandari and co-workers [2], [7], the experimental results validate that ultrasound radio-frequency (RF) signals can be faithfully reconstructed from modulo-wrapped samples by recovering the underlying fold sequence solely from the folded record y and threshold λ .

The physical operating limits of this particular implementation are governed by Itoh's condition (3.11) [2], [12], defined by the maximum consecutive sample difference relative to peak amplitude, $FF_{\text{bound}} = \frac{\max |\Delta g|}{\max |g|}$. Above this theoretical floor, the residual read margin m (Eq. (6.3)) determines the noise, quantisation error, and fold-amplitude jitter tolerance before exact integer fold recovery fails.

Within these operational bounds (and excluding quantisation and jitter) the peak signal amplitude recoverable beyond the ADC input rail scales as $1/FF$. At 30 dB channel SNR, the empirical threshold for exact recovery was $FF = 0.32$ for the point-target scanline and $FF = 0.35$ for the speckle-dominated scanlines, corresponding to effective dynamic range expansions of $3.1\times$ and $2.9\times$, respectively (Table 6.2).

When channel noise, quantisation, and fold-amplitude jitter were evaluated simultaneously, exact fold sequence recovery was achieved across all Monte Carlo realisations at $FF = 0.40$ for 8-bit and 10-bit quantisation, 20 dB channel SNR, and fold jitter severities up to 5% of 2λ . As illustrated by Figure 6.2, this operating point expands the usable dynamic range by $2.5\times$ relative to a conventional ADC operating with identical bit depth and rails at $\pm\lambda$.

At the system level, applying a shared threshold λ across receive elements models the physical input range constraints of real-world digitisers. Per-channel folding and recovery successfully reproduced the reference ideal B-mode image across operating conditions, failing on only 6 out of 8,256 receive channel recoveries (0.07%) at the harshest boundary condition ($FF = 0.35$, 20 dB SNR). At $FF = 0.35$, the reconstructed B-mode signals accommodate peak amplitudes reaching $2.9\times$ the native ADC threshold.

This work contributes by delivering an implementation of the USF for ultrasound RF processing and establishing its viability from individual receive channels through to delay-and-sum beamforming and B-mode image formation.

6.5 Limitations of the Obtained Results

Several constraints of this study highlight avenues for future development:

1. **Acoustic and Hardware Assumptions:** All simulations relied on a single phantom dataset acquired using a 5 MHz linear array transducer. Furthermore, the threshold λ was assumed to be known precisely; hardware miscalibration of λ remains an unaddressed source of error for a physical front end.
2. **Sampling and Algorithm Constraints:** The algorithm assumes temporal oversampling to satisfy Itoh's condition and relies on reading even fold counts. Sub-Nyquist sampling regimes and odd fold counts fall outside the scope of this implementation.
3. **Search Complexity under Severe Stressors:** When low SNR, coarse quantisation, or heavy fold jitter prevents a confident direct read, the algorithm falls back to a candidate-search mechanism. The search space grows rapidly as λ decreases or noise rises; above 2×10^4 candidates, the search is truncated to preserve moderate computational times. A more scalable fallback strategy is required for higher-noise and lower- λ regimes.
4. **Depth-Dependent Attenuation and Frequency Shift:** Tissue attenuation and time-gain compensation (TGC) were omitted, yielding flat amplitude profiles over depth. In real physical acquisitions, attenuation degrades SNR with depth, exposing deeper folds to higher relative noise, and induces a downward spectral shift in centre frequency, requiring dynamic tracking of both the search window and reserved harmonic count along the record.

6.6 Chapter Summary

We can summarise the following empirical findings:

1. At high channel SNR (80 dB), fold-sequence recovery of ultrasound RF is exact provided the folding threshold satisfies Itoh's condition ($FF > FF_{\text{bound}}$).
2. Exact recovery is maintained under simultaneous channel noise, coarse quantisation, and fold-amplitude jitter, provided the perturbation each introduces stays within the read margin $m = \lambda - \max_k |(\Delta g)[k]|$.

3. Applying the recovery procedure independently to receive channels under a uniform threshold λ successfully circumvents dynamic range clipping artefacts prior to beamforming.
4. Per-channel recovery faithfully reproduces the unconstrained benchmark B-mode image, expanding the range of possible recorded amplitudes by up to $2.9\times$ relative to λ .

In summary, fold-sequence recovery is exact as long as λ fulfils Itoh's condition. Moreover, recovery is still possible under noise, quantisation, and wrap jitter, as long as the read margin $m = \lambda - \max_k \|(\Delta g)[k]\|$ large enough. The same procedure on the receive channels, at one shared λ , reproduces the benchmark B-mode image by avoiding clipping.

7

Reflections

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7.1 Sustainability and societal impact

If that scheme discussed in this work were successfully carried into hardware and shown to add value, the modulo converter would replace the front-end converters of existing scanners, and the environmental cost of that replacement would have to be weighed against the benefit, as this would imply retiring functioning electronics before the end of their lifetime. The cost would only be justified if the gain in dynamic range translates into a diagnostic benefit that current hardware cannot provide. The work presented here does not settle whether that condition is met.

Conclusions and Future Work

This work set out to evaluate, in simulation, the operating regime over which an algorithm built on the Unlimited Sensing Framework (USF) reconstructs ultrasound radio-frequency (RF) signals from centred-modulo samples, and to quantify the resulting dynamic range extension. The research objectives outlined in Section 1.3.2 are addressed below.

First, to establish a practical recovery procedure using only the folded samples y and the threshold λ , Chapter 4 developed an algorithm based on the USF difference identity. The procedure estimates fold locations from sample differences, validates candidates against spectral content outside the transducer passband, and resolves ambiguous readings by searching candidate fold pairs constrained by pulse-echo wavelet symmetry. Across the three test scanlines (Chapter 5), under high channel SNR and ideal fold transitions, the algorithm achieved 100% exact fold-sequence recovery across all 50 Monte Carlo realisations down to the theoretical Itoh boundary, handling records containing up to 256 folds.

Second, to define the operational boundary of the method, Chapter 6 validated that performance is governed by Itoh’s condition [2], [12]. The remaining read margin (Eq. 6.3) determines the algorithm’s tolerance to channel noise, coarse quantisation, and fold-amplitude jitter. In high SNR settings, recoverable peak amplitudes reached approximately 3λ . Under combined non-ideal conditions (SNR = 20 dB, 8-bit quantisation, and 5% fold jitter at FF = 0.40), exact fold sequences were recovered across all trials. When integrated into a complete beamforming pipeline with a uniform threshold (FF = 0.35), per-channel recovery successfully reproduced the benchmark B-mode image in 40 dB regimes, with only 6 out of 8,256 channel recoveries failing at 20 dB SNR.

Limitations

The conclusions of this study are bounded by the scope defined in Section 1.3.3:

- **Acoustic and Hardware Modeling:** Simulations were restricted to a single phantom, one linear array transducer (5 MHz centre frequency), and a fixed sampling rate, without

accounting for tissue attenuation or time-gain compensation.

- **Calibration Constraints:** The fold threshold λ was assumed to be known precisely; the operational impact of hardware threshold miscalibration remains unexamined.
- **Residual Jitter Distortion:** While exact fold sequences are recovered in the presence of fold-amplitude jitter, amplitude reconstruction errors persist because the algorithm assumes discrete step heights of 2λ .
- **Oversampling Requirements:** The accuracy of direct-read recovery relies on high temporal oversampling, and performance has not yet been validated on clinical in vivo RF data.

Future Work

- Evaluate recovery performance across varying sampling rates.
- Implement post-recovery amplitude fitting to estimate exact wrap heights once fold locations are identified.
- Quantify algorithm sensitivity to miscalibration between the digitiser threshold and the reconstruction parameter λ .
- Implements realistic acoustic attenuation and center-frequency downshift with depth.
- Extend validation to diverse transducer geometries, array types, and transmit centre frequencies.
- Benchmark performance on experimental phantom acquisitions and clinical in vivo ultrasound RF datasets.
- Construct a physical single-channel modulo ADC hardware prototype to validate real-time front-end digitisation.

A

Short worked Example of the Pair Search

This appendix instantiates the sets $\mathcal{G}$, $\mathcal{C}$, and $\mathcal{Q}$ of Section 4.3 on a small numerical case, so that the distinction between the pair catalogue and the candidate set can be seen directly.

A

Consider a record in which the cue marking of (4.2) returns the gap indices

$$\mathcal{G} = \{ 10, 12, 50, 53 \},$$

with pair width bound $w_{\max} = 5$, and in which the direct read (4.8) counts $\widetilde{M} = 4$ folds, hence two closed pairs.

The catalogue $\mathcal{C}$

By (4.3), a pair $(p, p + w)$ is admissible when both ends lie in $\mathcal{G}$ and $1 \leq w \leq w_{\max}$. Only two of the six possible index couples qualify,

$$\mathcal{C} = \{ (10, 12), (50, 53) \},$$

since every other couple, for instance $(12, 50)$, exceeds the width bound. Each member of $\mathcal{C}$ is nothing more than a pair of positions: it carries no sign and makes no statement about the record

as a whole.

The candidate set $\mathcal{Q}$

By (4.4), a candidate is assembled by selecting $\widetilde{M}/2 = 2$ pairs from $\mathcal{C}$ and assigning each a sign s_i . With both catalogue pairs required, the free choices are the two signs, and $\mathcal{Q}$ contains four candidate fold sequences. Each is a full length- $(K-1)$ vector that vanishes outside the four selected gaps:

Table A.1: The four members of $\mathcal{Q}$ for the example record. Each row is one complete candidate $\mathbf{q}'$; all other entries are zero.

Candidate	$q'[10]$	$q'[12]$	$q'[50]$	$q'[53]$	Interpretation
$\mathbf{q}_1$	+1	-1	+1	-1	both echoes exit through $+\lambda$
$\mathbf{q}_2$	+1	-1	-1	+1	first exits upward, second downward
$\mathbf{q}_3$	-1	+1	+1	-1	first downward, second upward
$\mathbf{q}_4$	-1	+1	-1	+1	both exit through $-\lambda$

In the notation of (4.4), the first row is $\mathbf{q}_1 = (\mathbf{e}_{10} - \mathbf{e}_{12}) + (\mathbf{e}_{50} - \mathbf{e}_{53})$, with both signs $s_1 = s_2 = +1$; the remaining rows differ only in the signs.

Every row of Table A.1 sums to zero. This is the closure property of (4.4): each pair contributes one positive and one negative unit jump, so the staircase returns to its starting level after every excursion.

Selection

The misfit (4.7) is evaluated on complete candidates only; a lone catalogue pair cannot be scored. Here the search would compare $\rho(\mathbf{q}_1), \dots, \rho(\mathbf{q}_4)$ and retain the smallest, since the correct sign pattern cancels the fold signature in the out-of-band bins and leaves only noise. In practice the search is not always reached: if the direct read returns, say, +1 at gap 10, -1 at gap 12, +1 at gap 50, and -1 at gap 53, the seed pairing of Algorithm 1 assembles $\mathbf{q}_1$ immediately, and the verification step confirms it.

On a measured record, the dilation in (4.2) marks whole neighbourhoods rather than isolated gaps, so $\mathcal{C}$ holds hundreds of pairs and the size of $\mathcal{Q}$ grows combinatorially with the fold count.

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